

 Weatherford®		PHOTO DENSITY DUAL NEUTRON	
COMPANY S.N.G.N. ROMGAZ S.A.			
WELL 209 NADES			
FIELD NADES			
PROVINCE/COUNTY MURES			
COUNTRY/STATE ROMANIA			
LOCATION X = 529925.187 m Y = 477271.675 m			
FIELD PRINT			
Latitude	46°16'07.03" N	Other Services	
Longitude	24°42'12.69" E	DUAL LATEROLOG MICROLATEROLOG BOREHOLE NAVIGATION	
Permanent Datum GL, Elevation 480.21 metres		XY CALIPER COMPENSATED SONIC	
Log Measured From KB, 4.50 metres above Permanent Datum		Elevations: KB 484.71 DF 484.71 GL 480.21	
Drilling Measured From KB			
Date	04-SEP-2024		
Run Number	2B		
Service Order	24220368		
Depth Driller	1225.00	metres	
Depth Logger	1226.00	metres	
First Reading	1225.00	metres	
Last Reading	250.00	metres	
Casing Driller	250.00	metres	
Casing Logger	249.50	metres	
Bit Size	8.500	inches	
Hole Fluid Type	INHIBITIV		
Density / Viscosity	1.10 g/cc	50.00 SEC/L	
PH / Fluid Loss	9.50	4.10 ml/30Min	
Sample Source	FLOWLINE		
Rm @ Measured Temp	0.15 @ 25.0	ohm-m	
Rmf @ Measured Temp	0.12 @ 25.0	ohm-m	
Rmc @ Measured Temp	0.18 @ 25.0	ohm-m	
Source Rmf / Rmc	PRESS	FILTER	
Rm @ BHT	0.11 @ 39.0	ohm-m	
Time Since Circulation	6 HOURS		
Max Recorded Temp	39.00	deg C	
Equipment / Base	13347	PLO	
Recorded By	MALEA CRISTIAN		
Witnessed By	VASILE FARCAS		

In interpreting, communicating or providing information and/or making recommendations, either written or oral, as to logs or test or other data, type or amount of material, or Work or other service to be furnished, or manner of performance, or in predicting results to be obtained, the Contractor will give the Company the benefit of the Contractor's best judgment based on its experience and will perform all such Work in a good and workmanlike manner. Any interpretation of test or other data, and any recommendation or reservoir description based upon such interpretations, are opinions based upon inferences from measurements and empirical relationships and assumptions, which inferences and assumptions are not infallible, and with respect to which professional engineers and analysts may differ. ACCORDINGLY ANY INTERPRETATION OR RECOMMENDATION RESULTING FROM THE SERVICES WILL BE AT THE SOLE RISK OF THE COMPANY, AND THE CONTRACTOR CANNOT AND DOES NOT WARRANT THE ACCURACY, CORRECTNESS OR COMPLETENESS OF ANY SUCH INTERPRETATION OR RECOMMENDATION, WHICH INTERPRETATIONS AND RECOMMENDATIONS SHOULD NOT, THEREFORE, UNDER ANY CIRCUMSTANCES BE RELIED UPON AS THE SOLE OR MAIN BASIS FOR ANY DRILLING, COMPLETION, WELL TREATMENT, PRODUCTION OR FINANCIAL DECISION, OR ANY PROCEDURE INVOLVING ANY RISK TO THE SAFETY OF ANY DRILLING ACTIVITY, DRILLING RIG OR ITS CREW OR ANY OTHER INDIVIDUAL. THE COMPANY HAS FULL RESPONSIBILITY FOR ALL DECISIONS CONCERNING THE SERVICES.

BOREHOLE RECORD					Last Edited: 04-SEP-2024 10:04
Bit Size inches		Depth From metres		Depth To metres	
12.250		23.00		246.00	
8.500		250.00		1225.00	
CASING RECORD					
Type	Size inches	Depth From metres	Shoe Depth metres	Weight pounds/ft	
SURFACE	16.000	0.00	23.00	0.00	
CASING	9.625	0.00	250.00	43.50	

REMARKS

Clients intervals and scales logged

Acquisition software: Weatherford Well Manager, version 24.03.1279

Cable used: CAMESA size 15/32", type 7J46RXZ with cable head and 8k weakpoint set

Depth wheel calibration factor used 0.6096

Depth Control Procedure for Multiple runs in the well followed.

Run2A: CBHC, MTAK, MBE, MBE, SHA, MCG, MBN, MTC-Y, MSS, MTC-X, SKJ, MUG, MLE, MLG, MTI, MGS,HFS

Run2B: CBHC, MTAK, SHA, MCG, SKJ, MMR, MDN, MPD, HFS

Run2C: CBHC, MTAK, SHA, MCG, MISD, SKJ, MISE, SKJ, MISD, MSS, MISD, HFS logged in combination.

Rm, Rmf, Rmc measured on samples from Mud Engineer given on 04.09.2024

Caliper measurements and averages from TD to casing:

Total hole volume = 35.32 cubic meters

Annular volume = 20.52 cubic meters, based on future casing size of 5.5 inches

Average hole diameter = 8.49 inches (215.77 mm), based on XY caliper

AFTER SURVEY CHECKS					
C:\Users\sysadmin\AppData\Local\Temp\Weatherford PreView8e\0\209_NADES_NUC_CALB.dta					
Gamma Check MCG-E.A 531			Field Calibration on 04-SEP-2024 05:54 After Survey Check on 04-SEP-2024 14:38		
	Before (API)		After (API)		
Background	48		66		
Calibrator (Gross)	755		773		
Calibrator (Net)	707		707		
Gamma Check Tolerances MCG-E.A 531					
Ratio	1.435	1.40 1.475 1.55	Counts/API		
Neutron Check MDN-C.A 528			Before Survey Check on 04-SEP-2024 06:06 After Survey Check on 04-SEP-2024 14:19		
	Near (cps)		Far (cps)		
	Before	After	Before	After	
	2325	2308	3371	3364	
	Ratio				
	Before		After		
	0.690		0.686		
Neutron Check Tolerances MDN-C.A 528					
After Check	0.686	0.670 0.690 0.710			
Photo Density Check MPD-D.A 425			Before Survey Check on 04-SEP-2024 06:02 After Survey Check on 04-SEP-2024 14:15		
Density Check					
	Near		Far		
	Before	After	Before	After	
	939.1	938.4	1084.0	1082.9	
PE Check					
	Before		After		
WS	177.0		178.0		
WH	825.4		825.0		
Photo Density Check Tolerances MPD-D.A 425					
Near Den. After Check	938.4	-3% 939.1 +3%	Far Den. After Check	1082.9	-3% 1084.0 +3%
PE WS After Check	178.0	-6% 177.0 +6%	PE WH After Check	825.0	-6% 825.4 +6%
Caliper Check MPD-D.A 425			Field Calibration on 04-SEP-2024 05:57 After Survey Check on 04-SEP-2024 13:36		

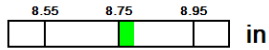
Before (in)
7.96

After (in)
8.79

Caliper Check Tolerances MPD-D.A 425

Short Arm After Check

8.79



MAIN PASS 1:1000



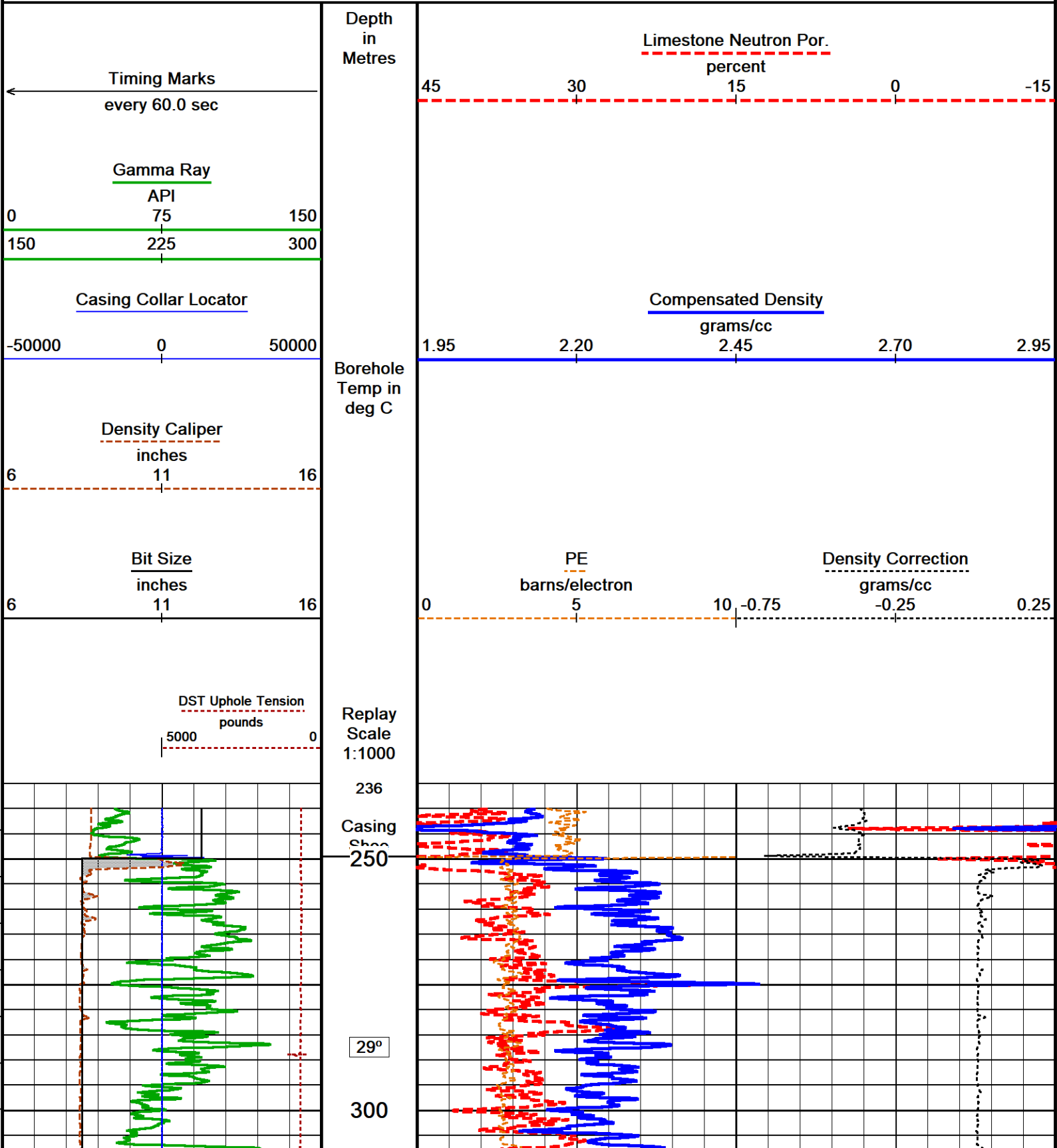
Depth Based Data - Maximum Sampling Increment 10.0cm

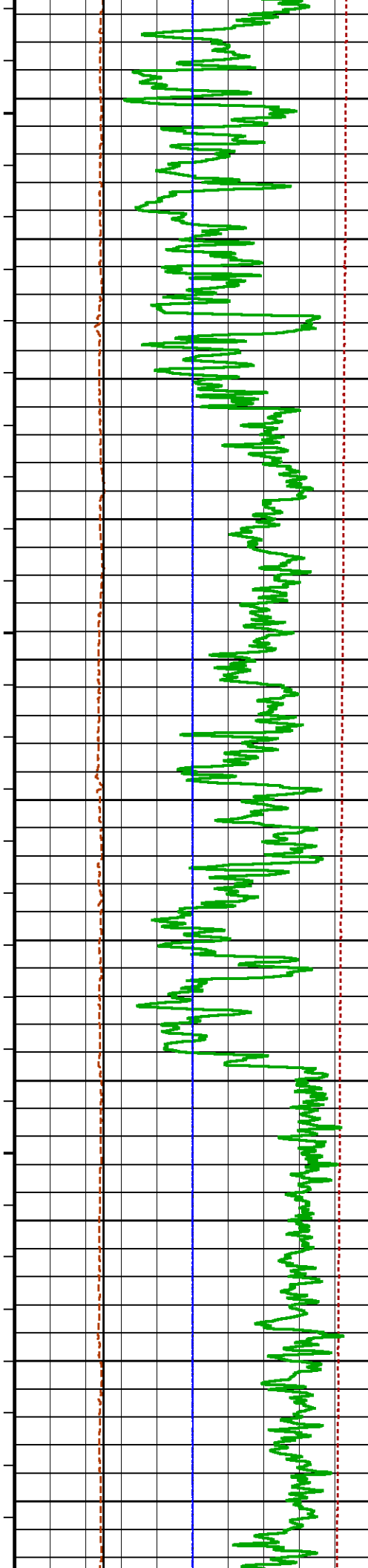
Plotted on 04-SEP-2024 16:28

Filename: C:\Users\systadmin\AppData\Local\Temp\Weatherford PreView...\209_NADES_NUC_MP.dta

Recorded on 04-SEP-2024 11:42

System Versions: Logged with 24.03.1279 Plotted with 22.01.1692





30°

350

30°

400

31°

450

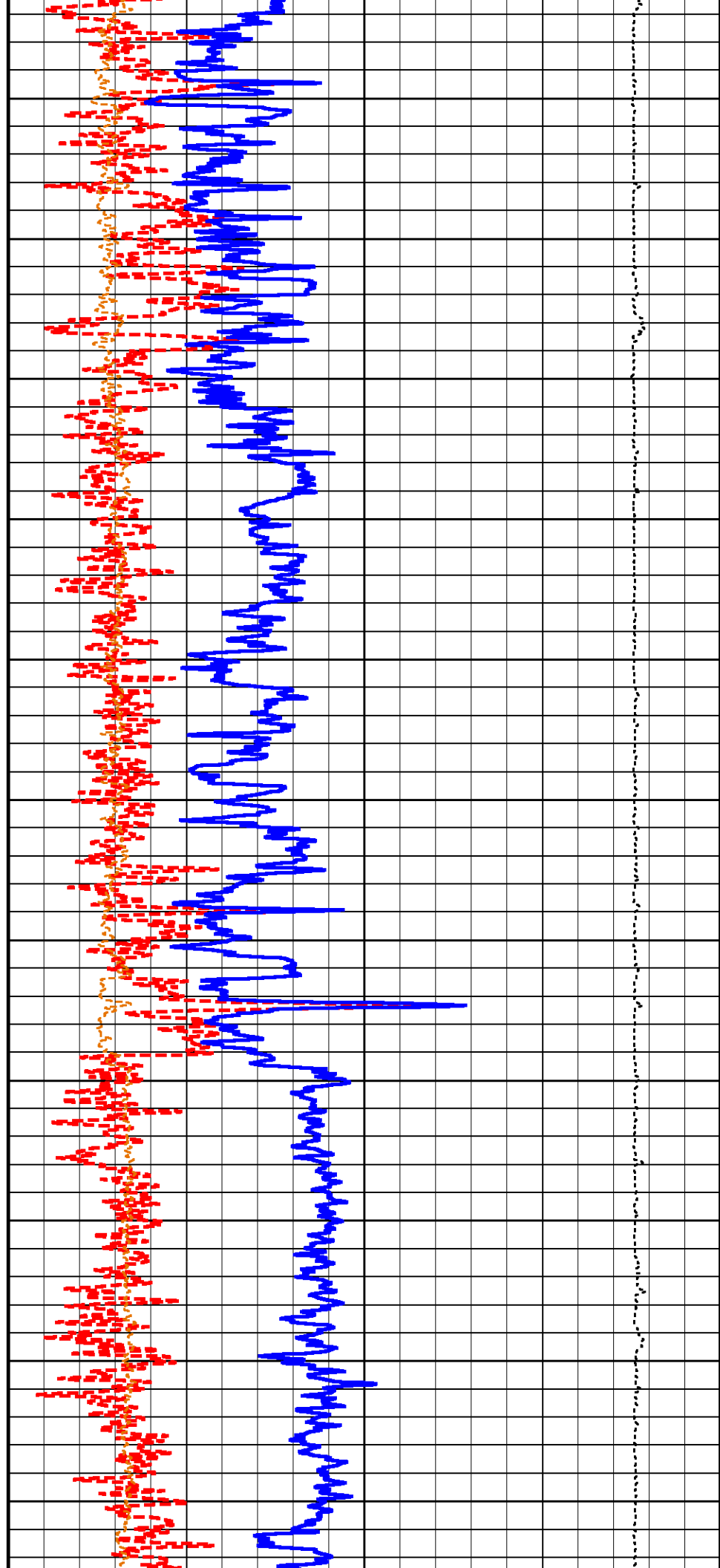
31°

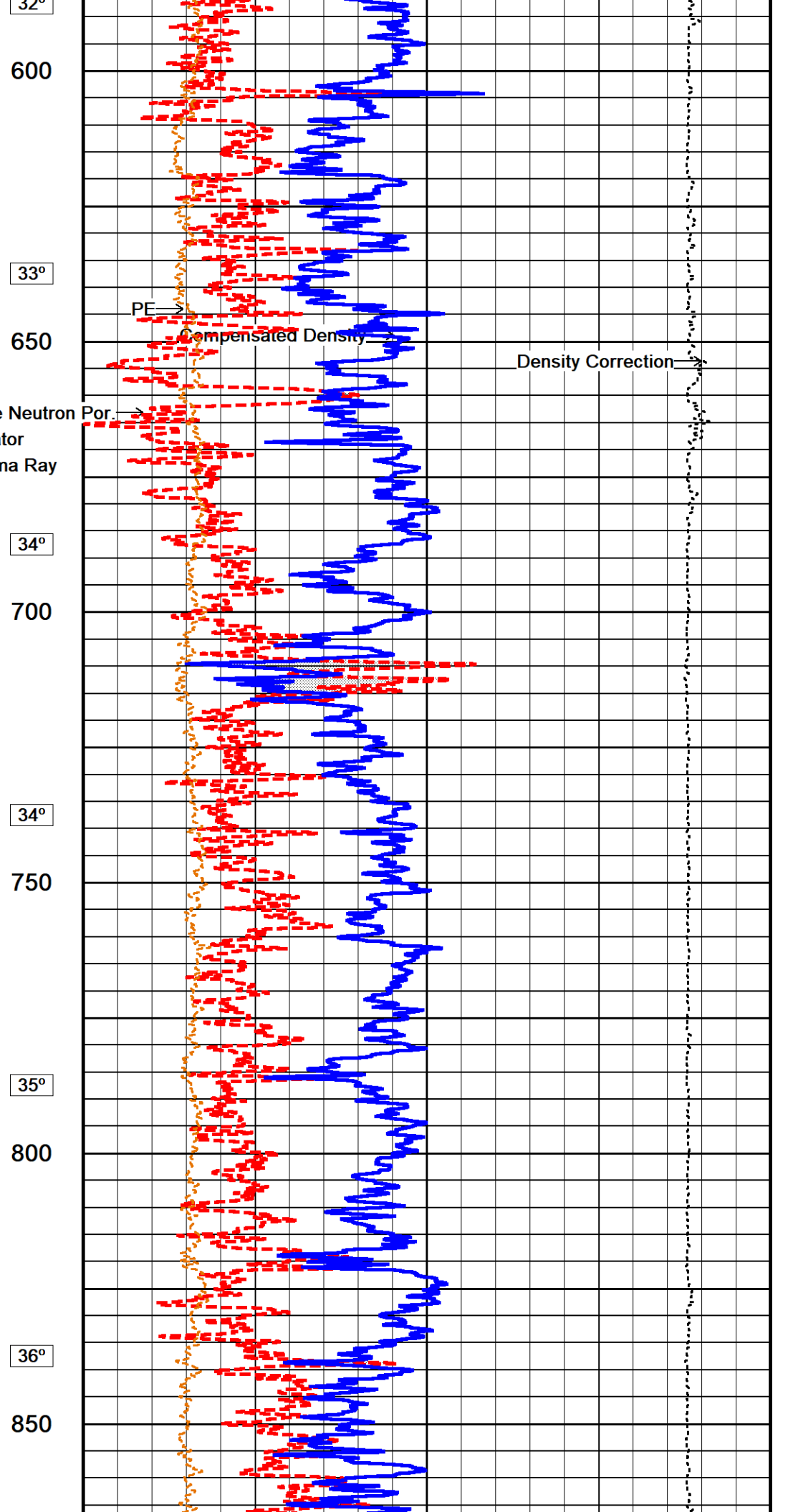
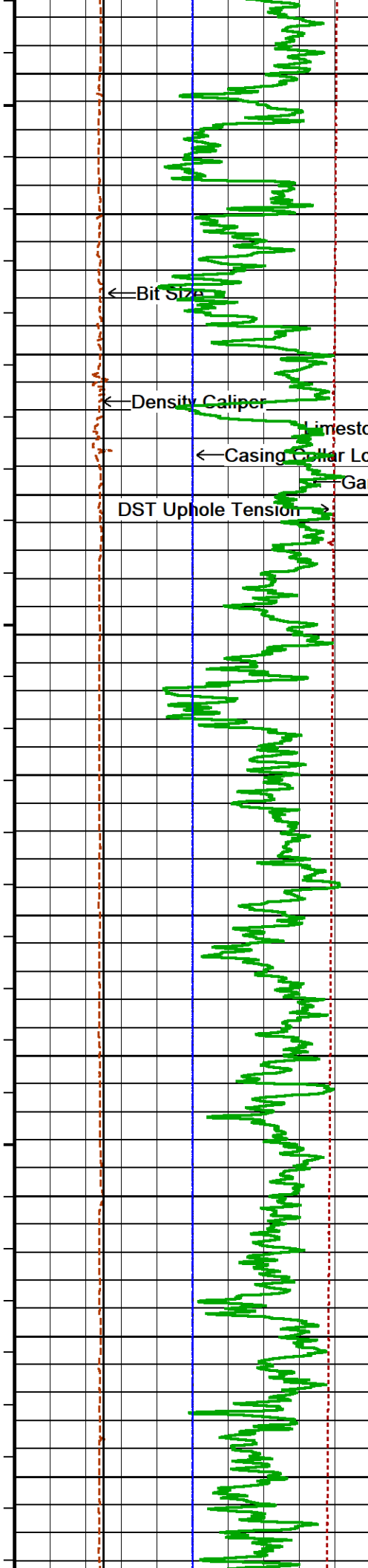
500

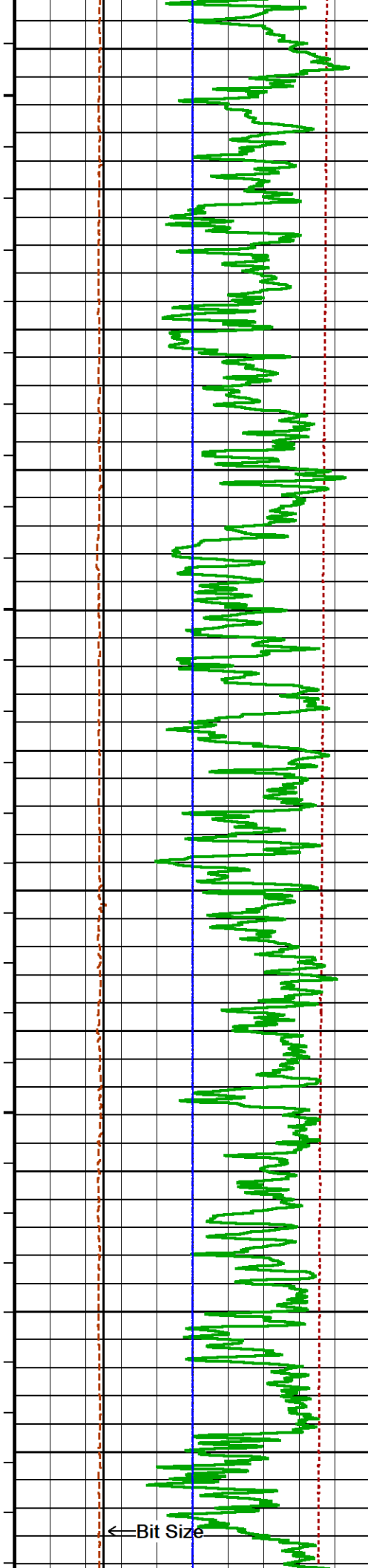
32°

550

33°







← Bit Size

36°

900

37°

950

37°

1000

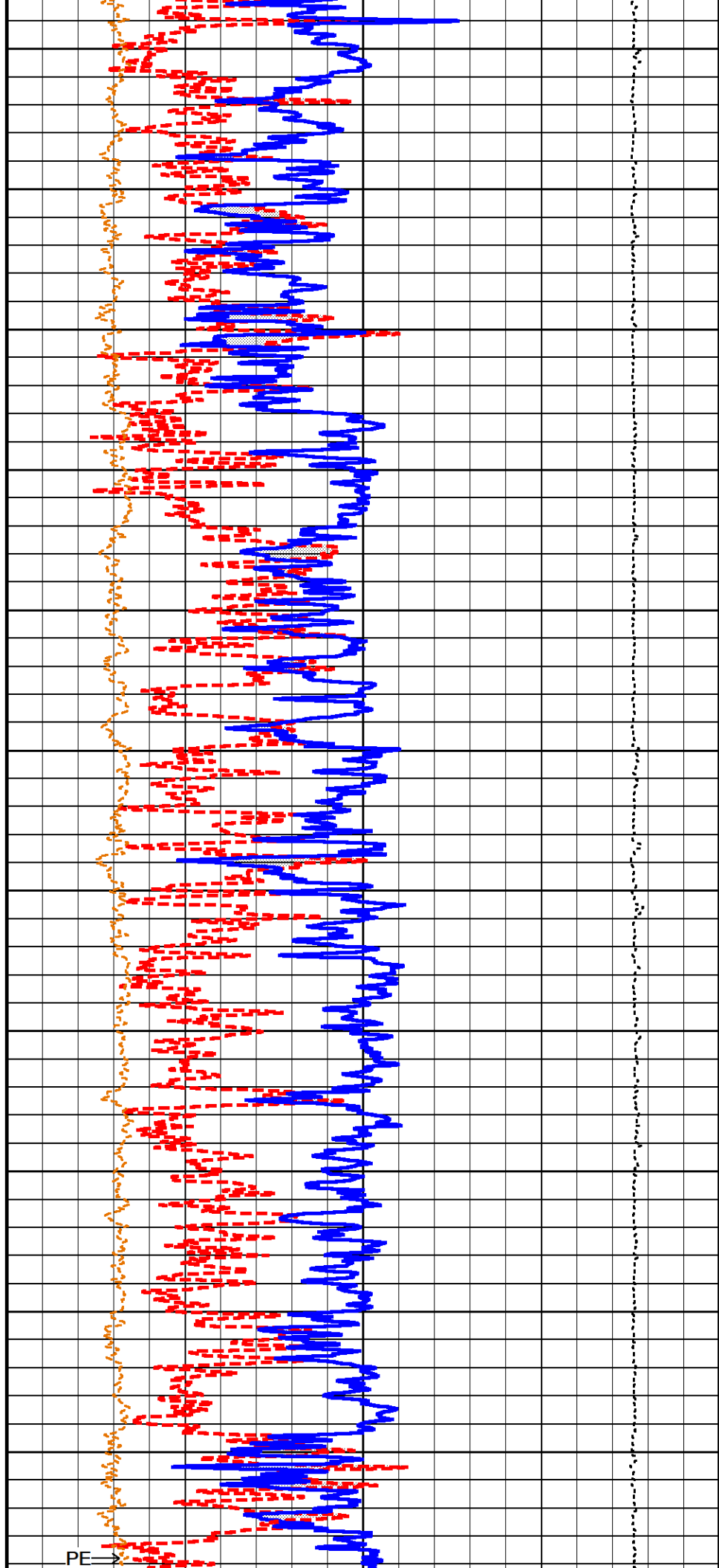
38°

1050

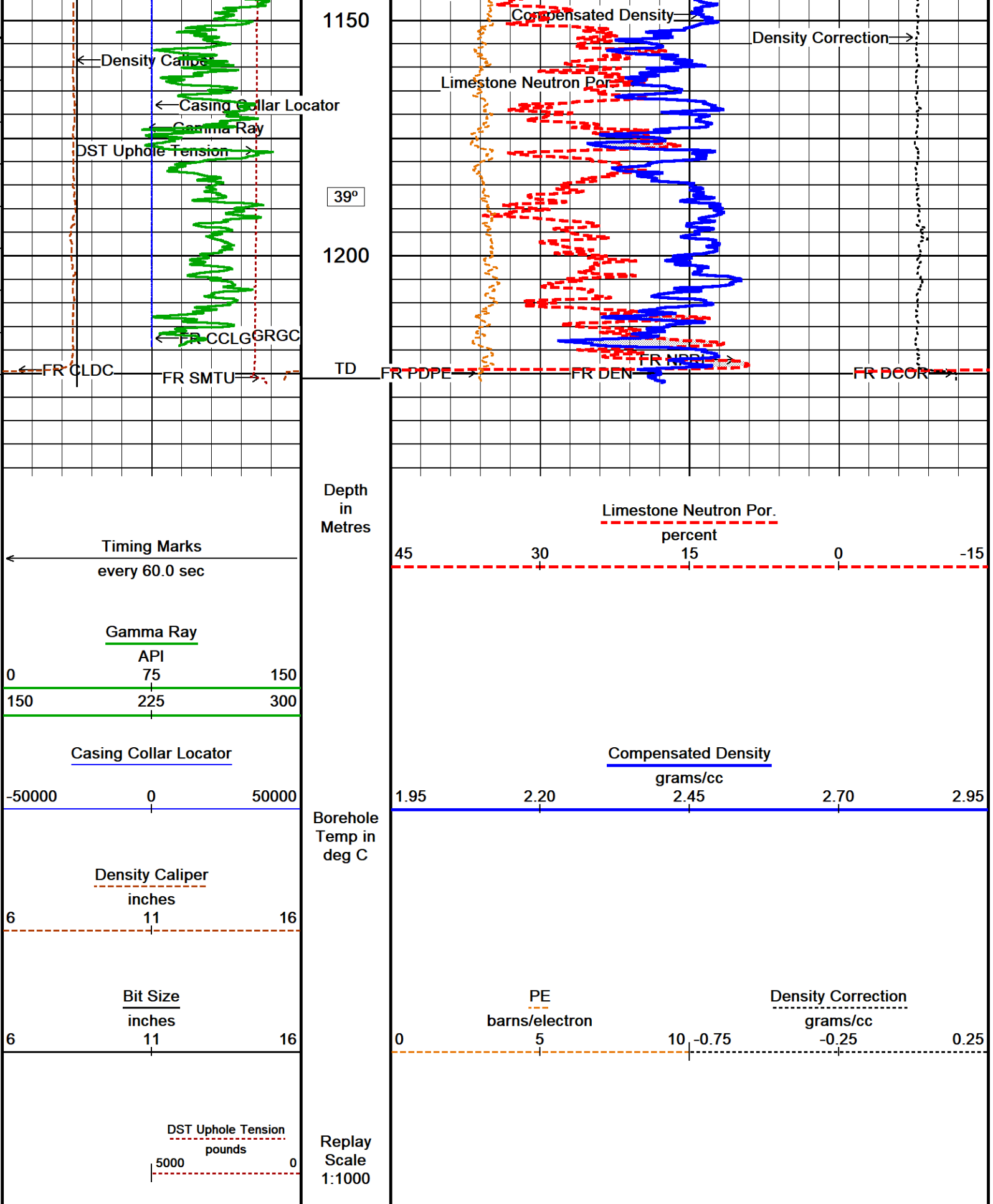
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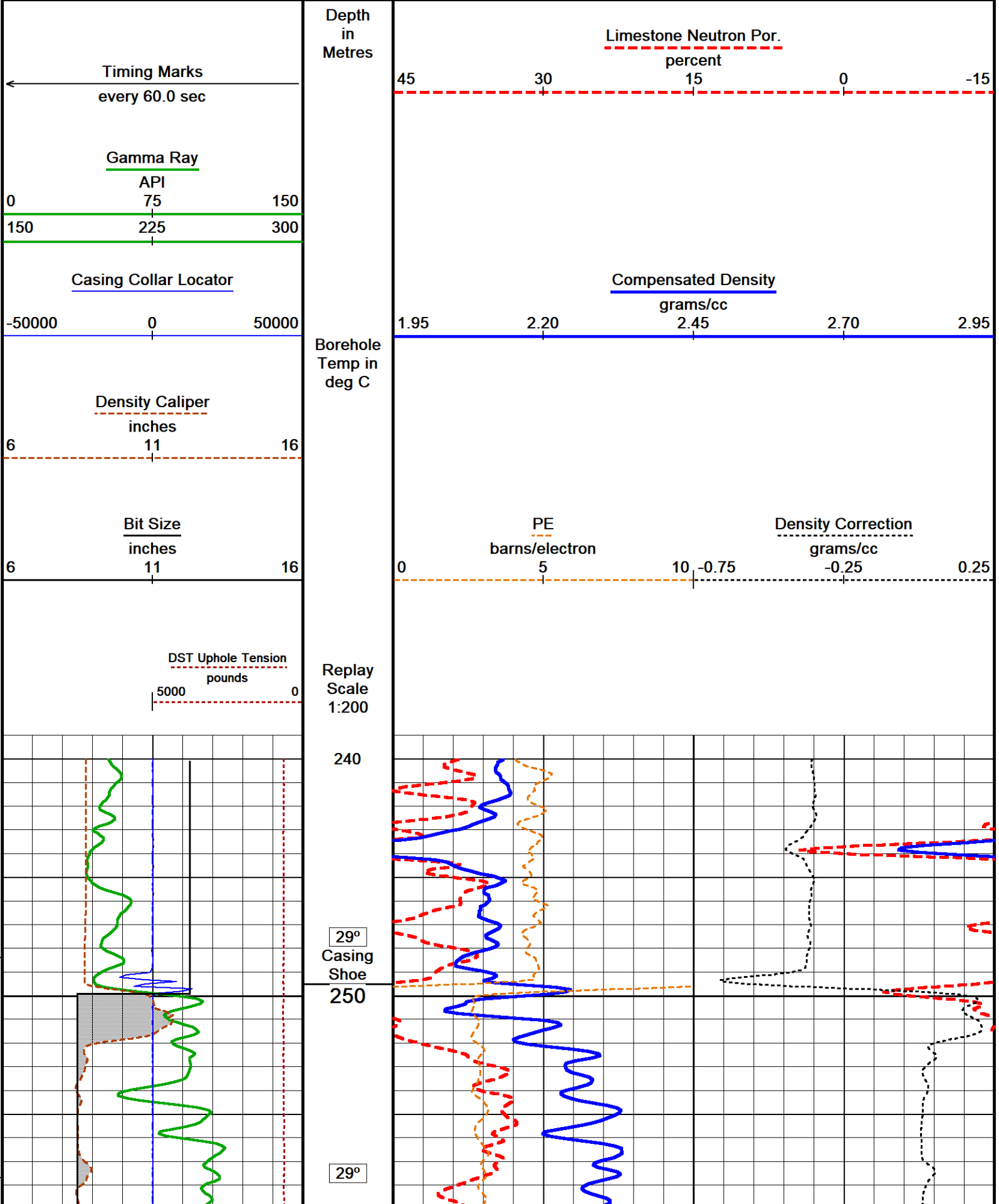
1100

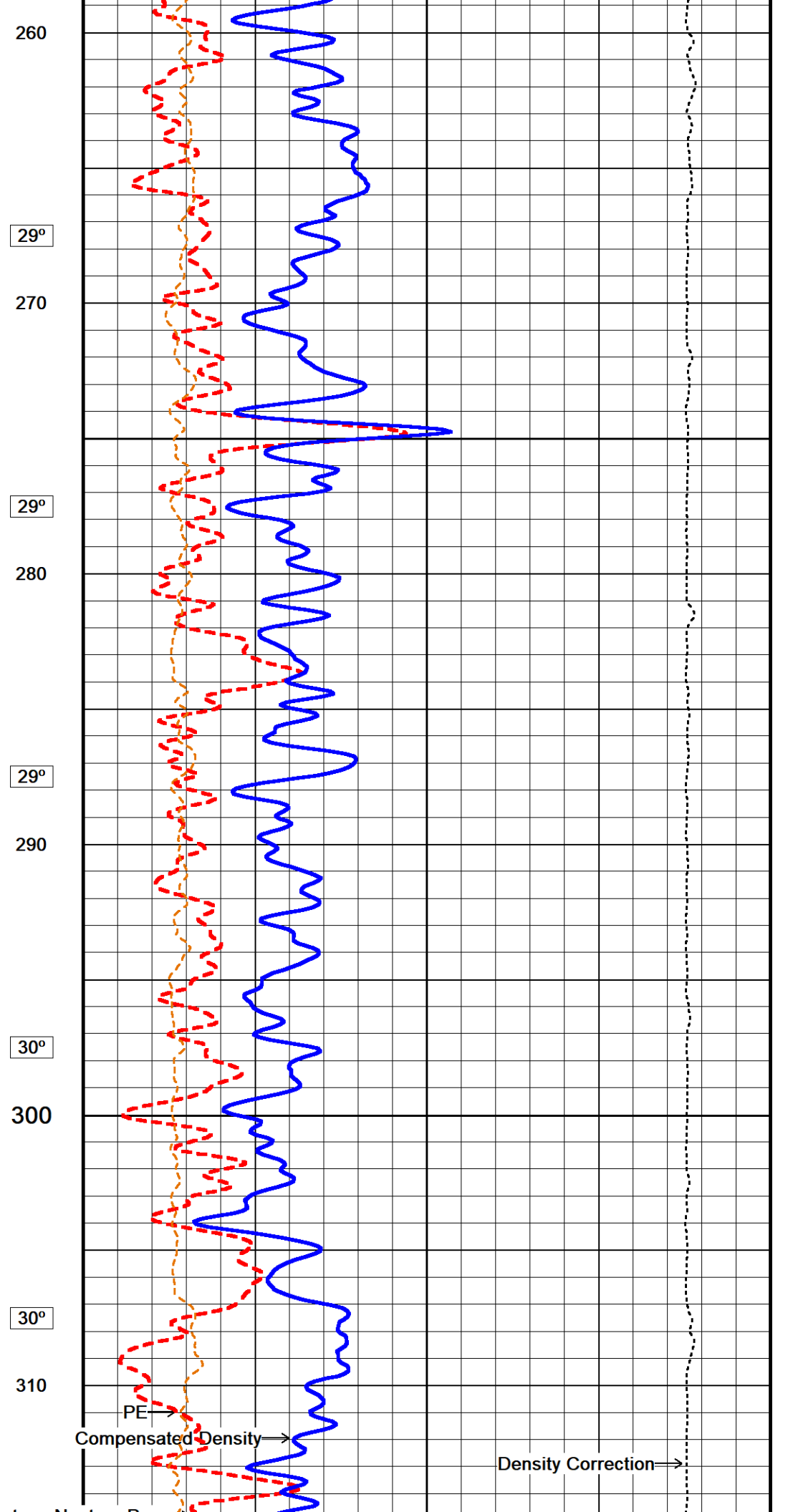
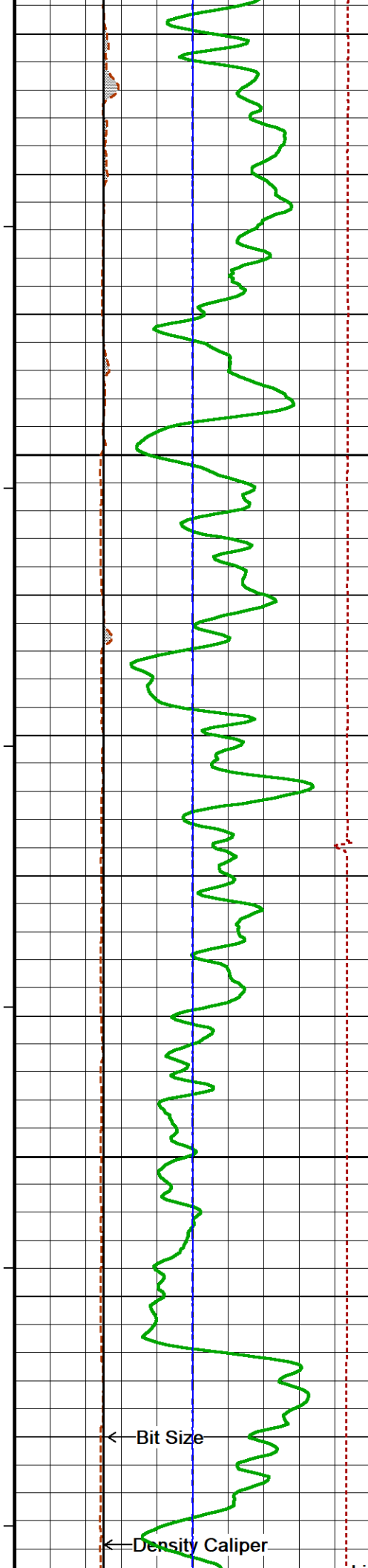
39°

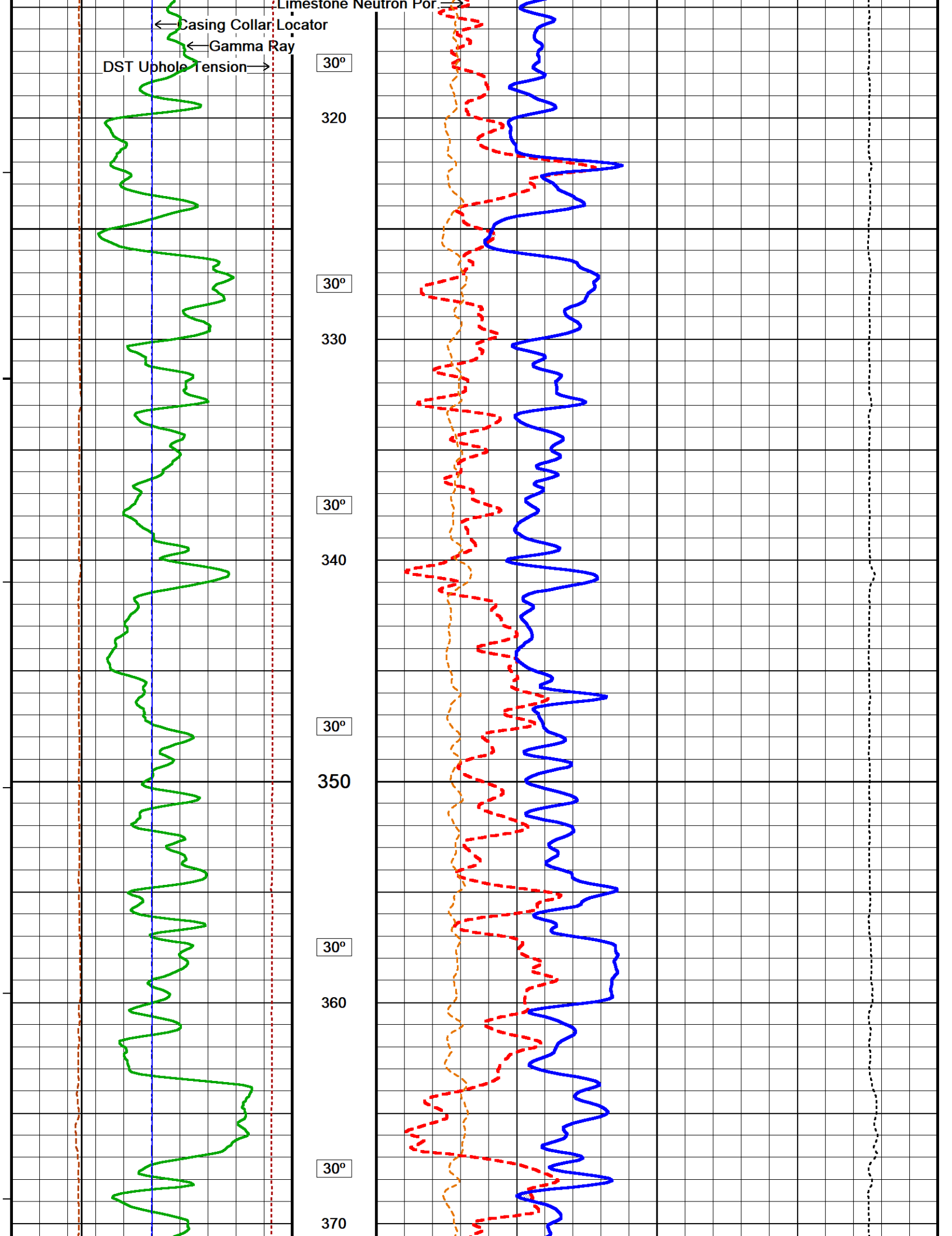


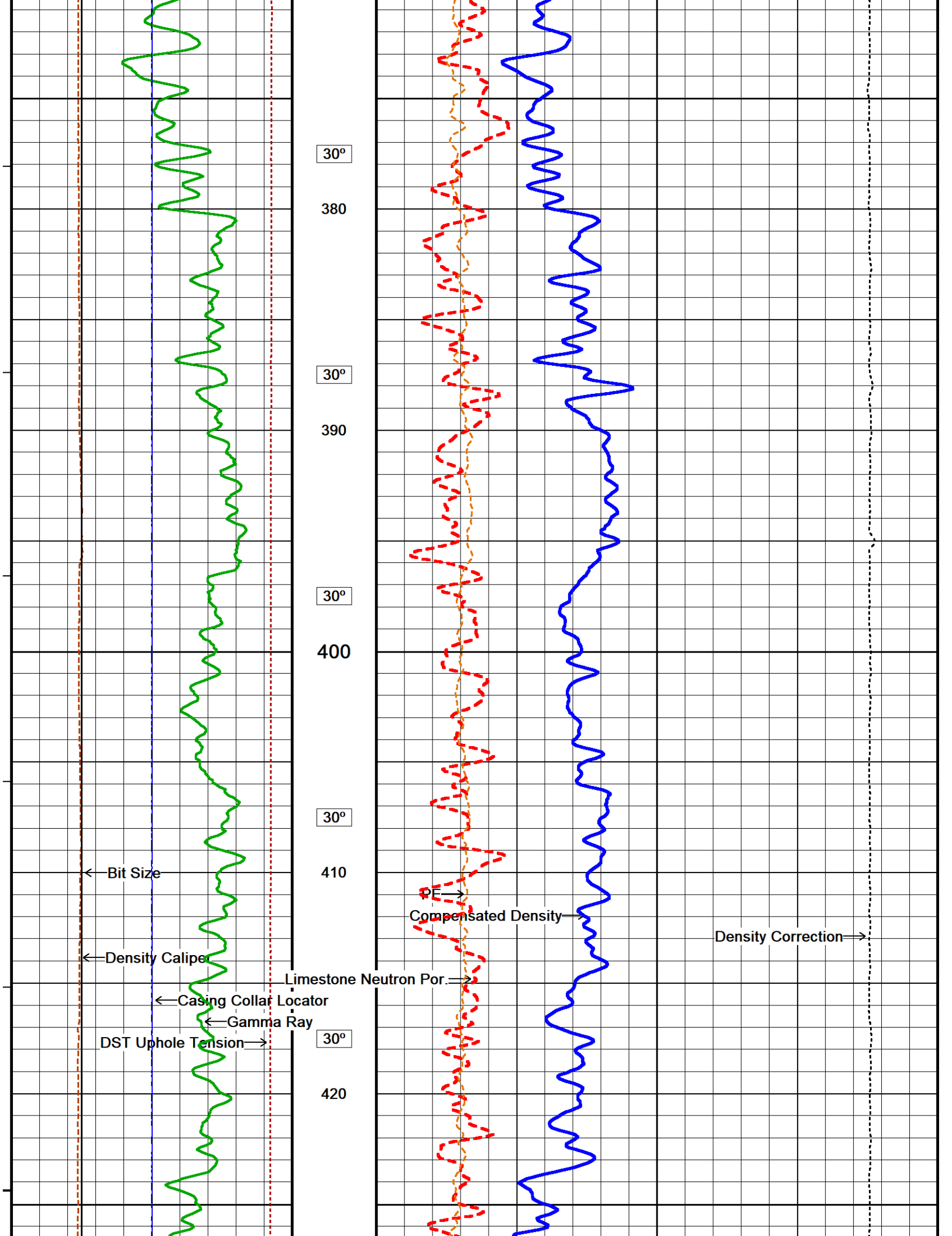
PE→

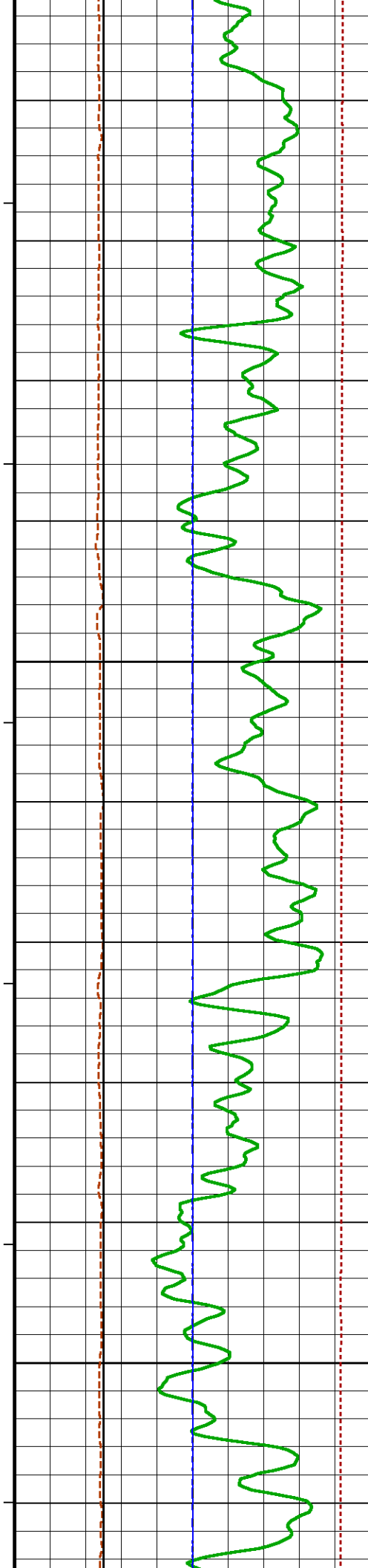












30°

430

31°

440

31°

450

31°

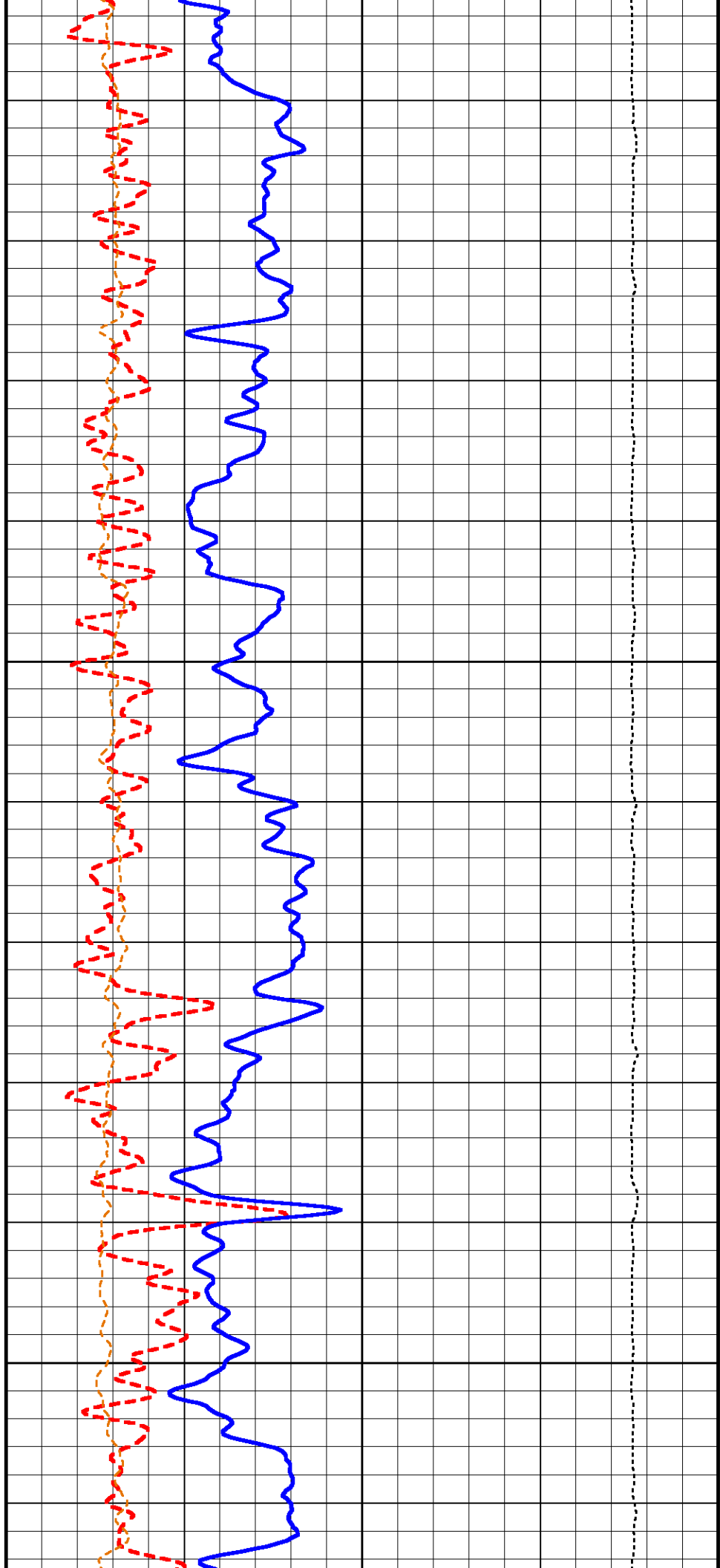
460

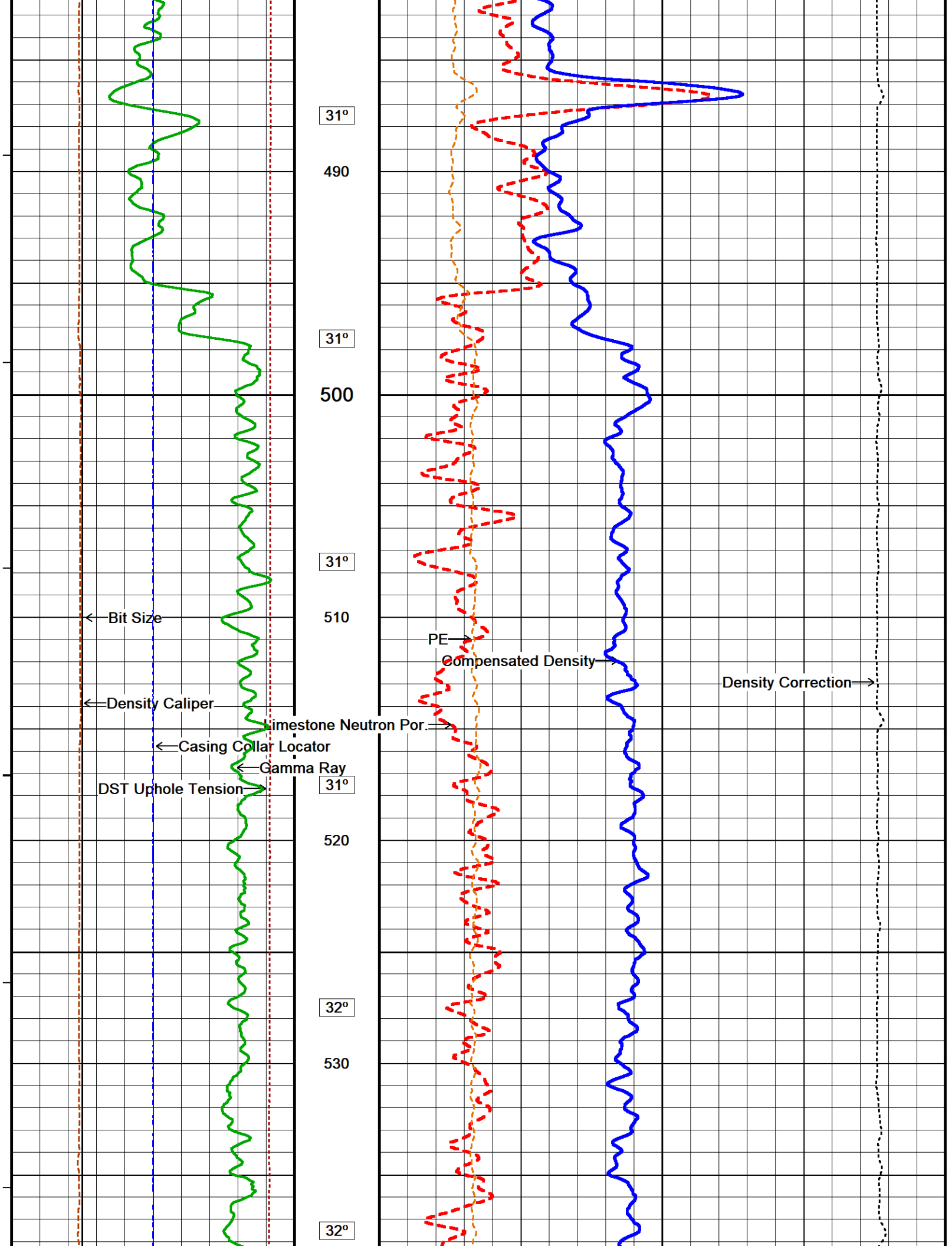
31°

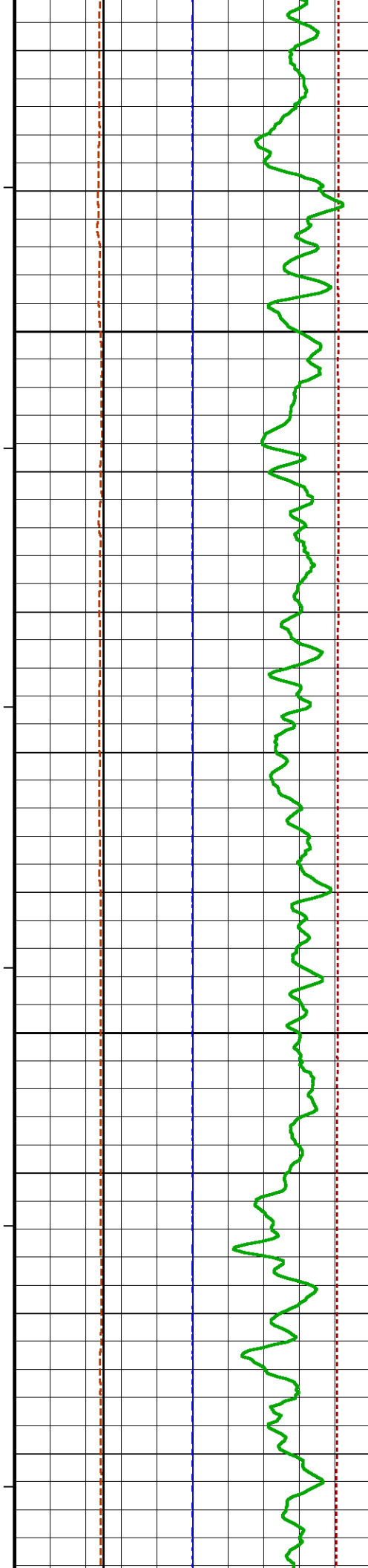
470

31°

480







540

32°

550

32°

560

32°

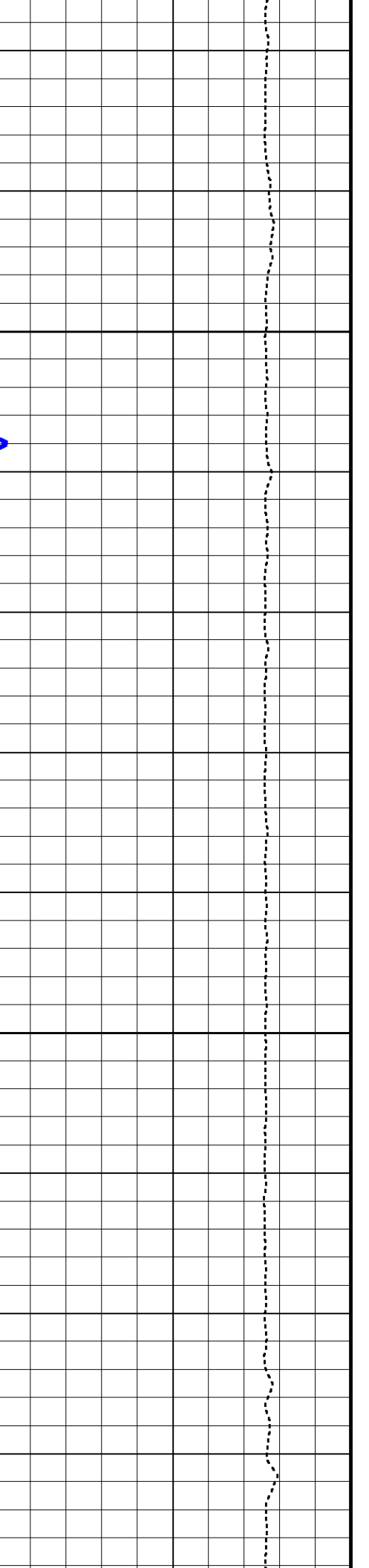
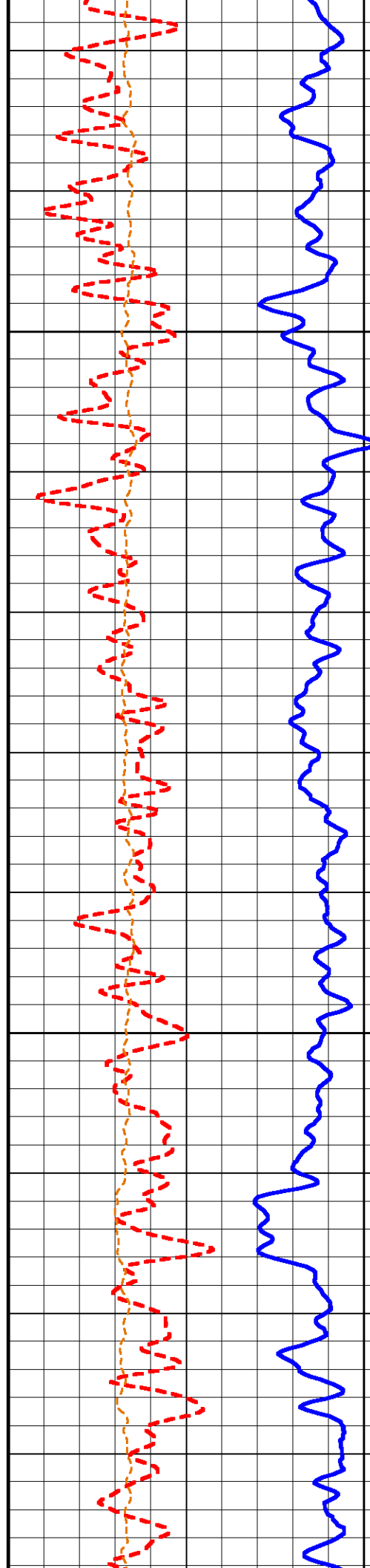
570

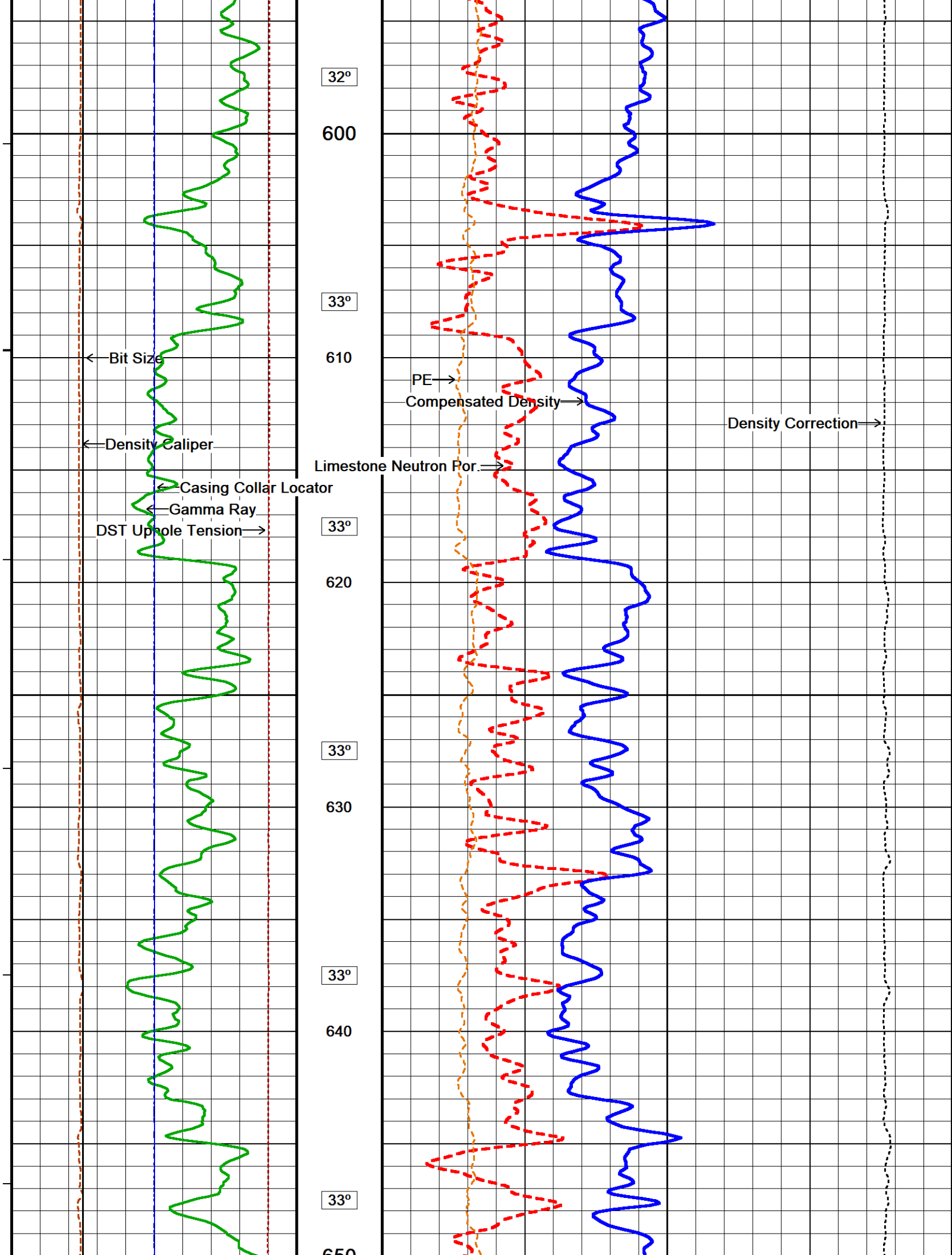
32°

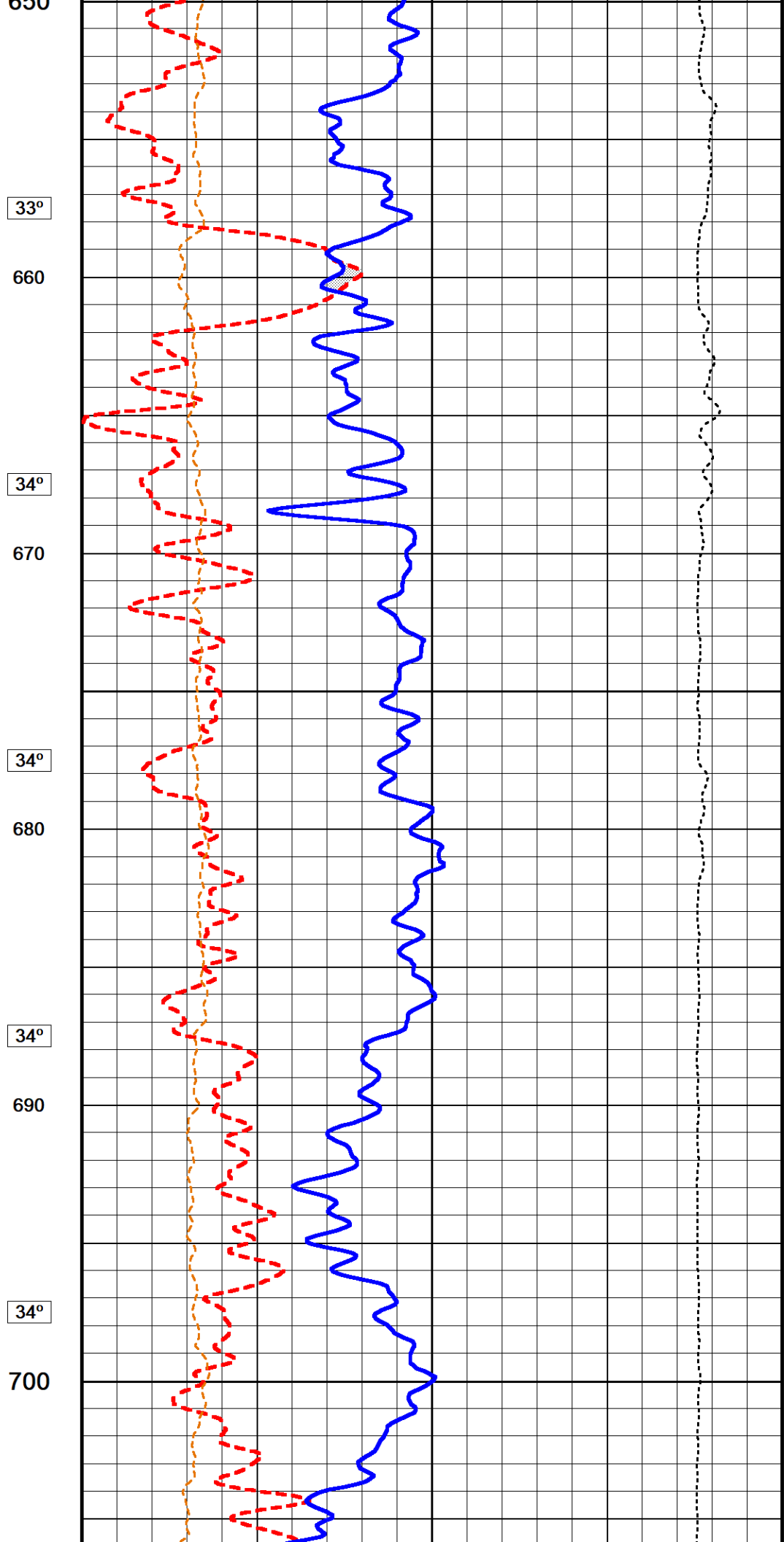
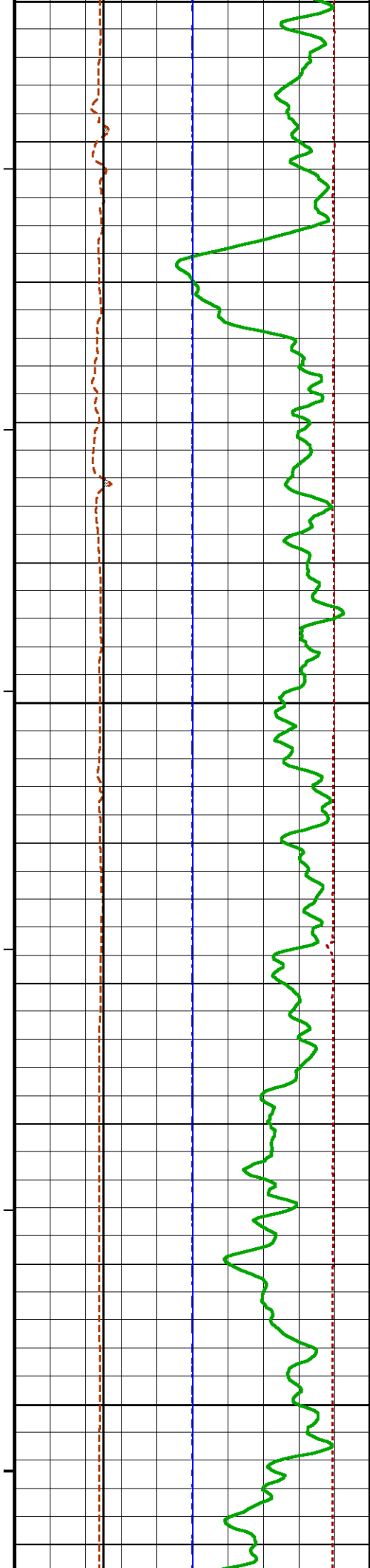
580

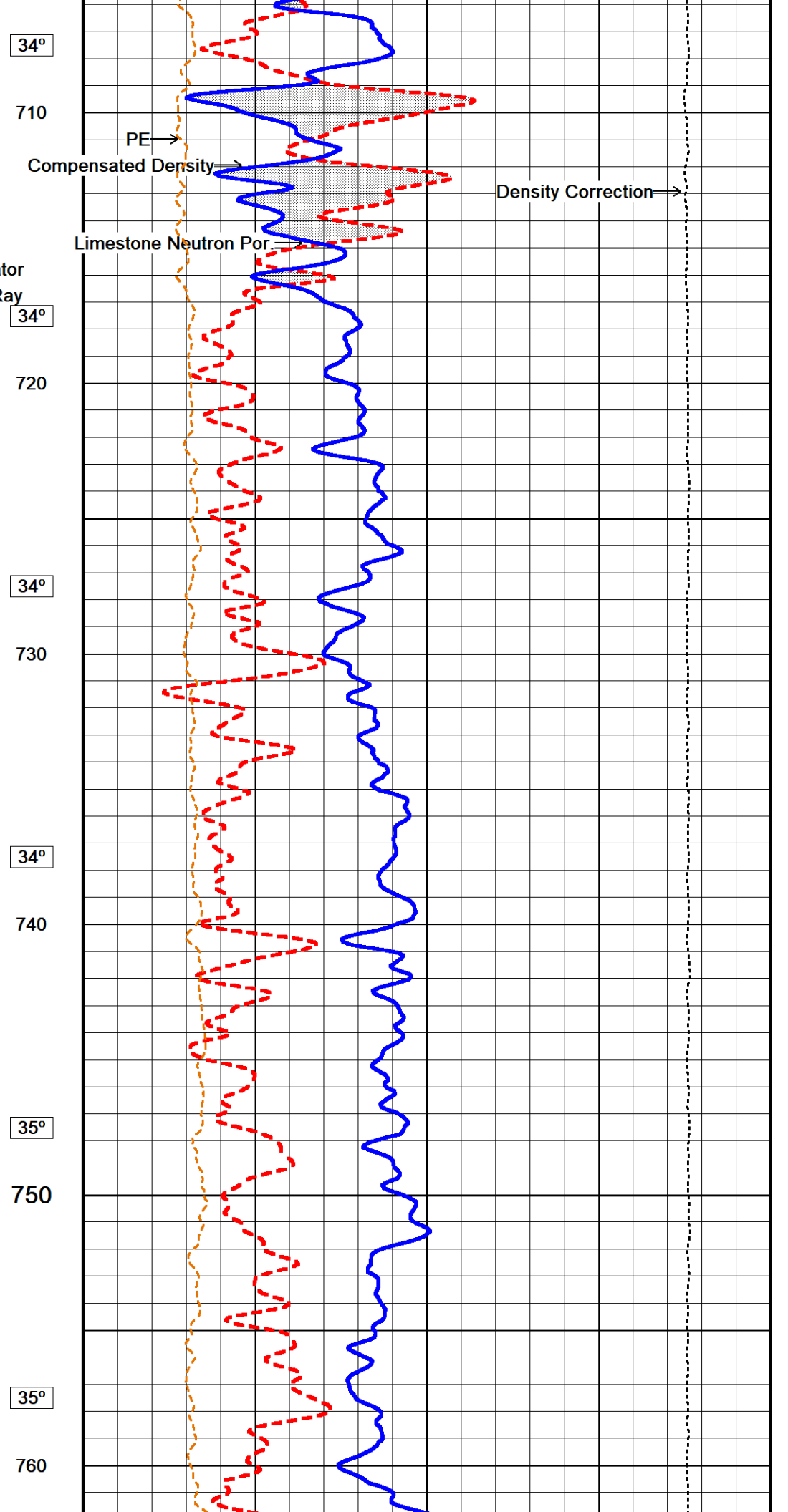
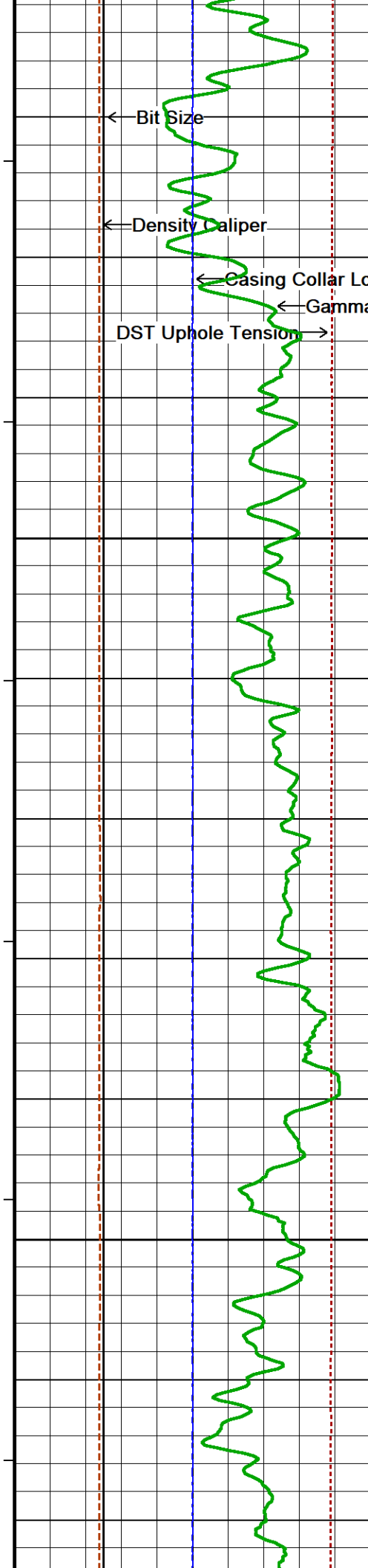
32°

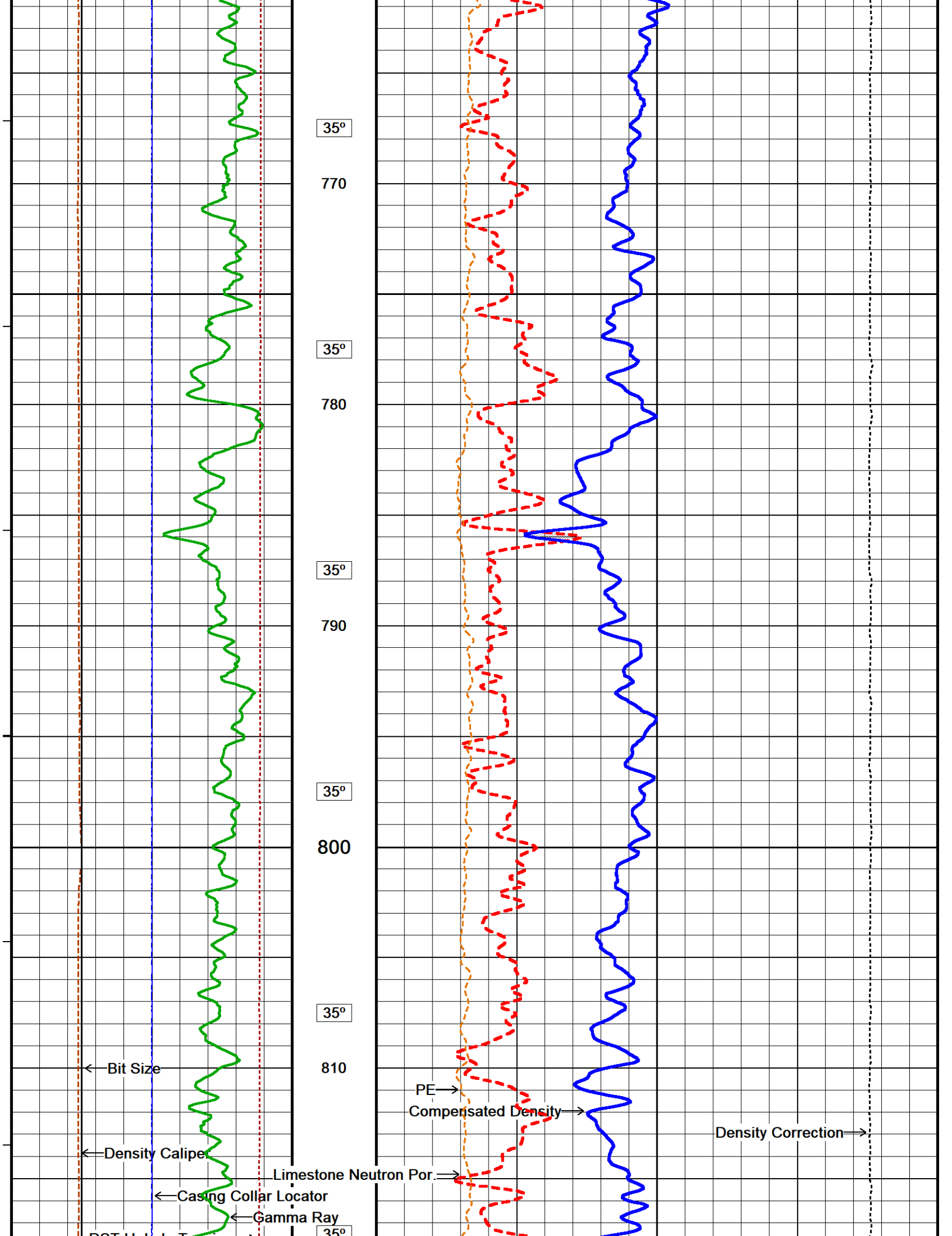
590

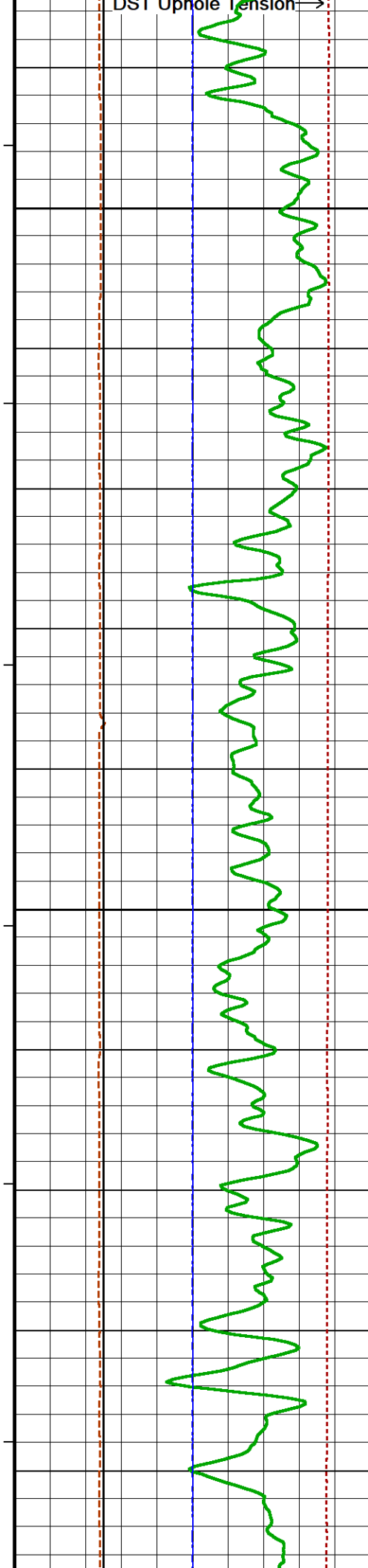




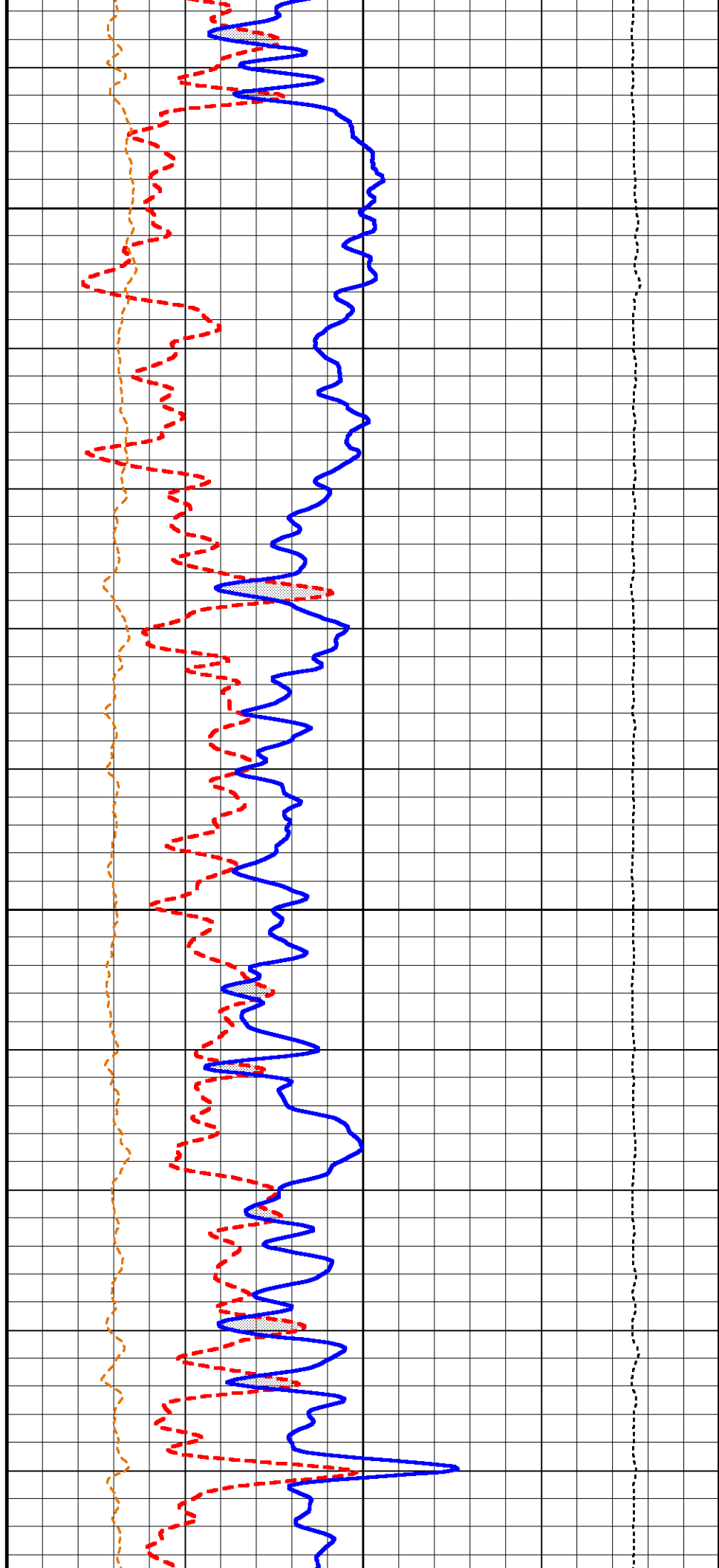


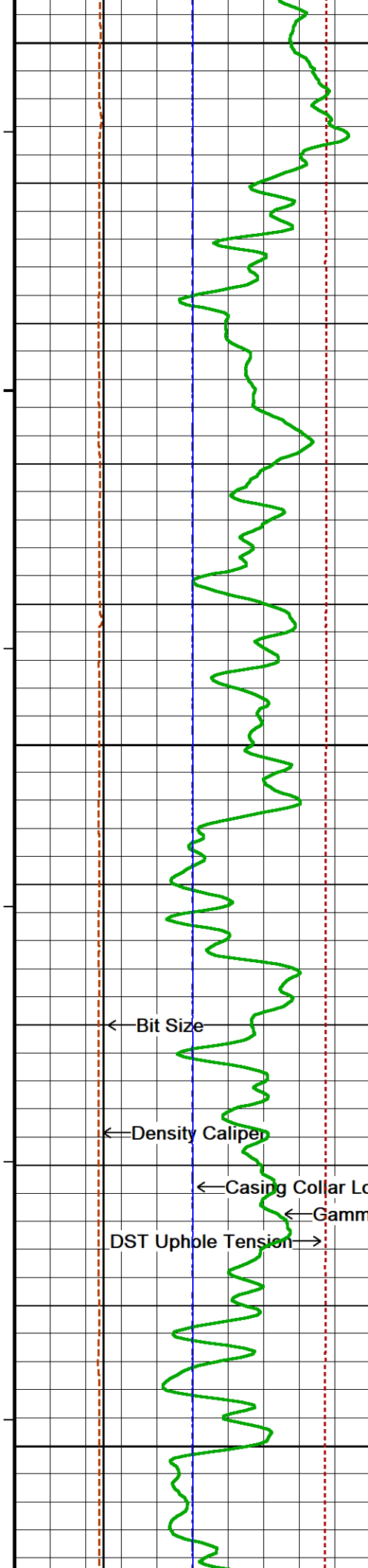






820
35°
830
36°
840
36°
850
36°
860
36°
870





36°

880

36°

890

36°

900

36°

910

36°

920

37°

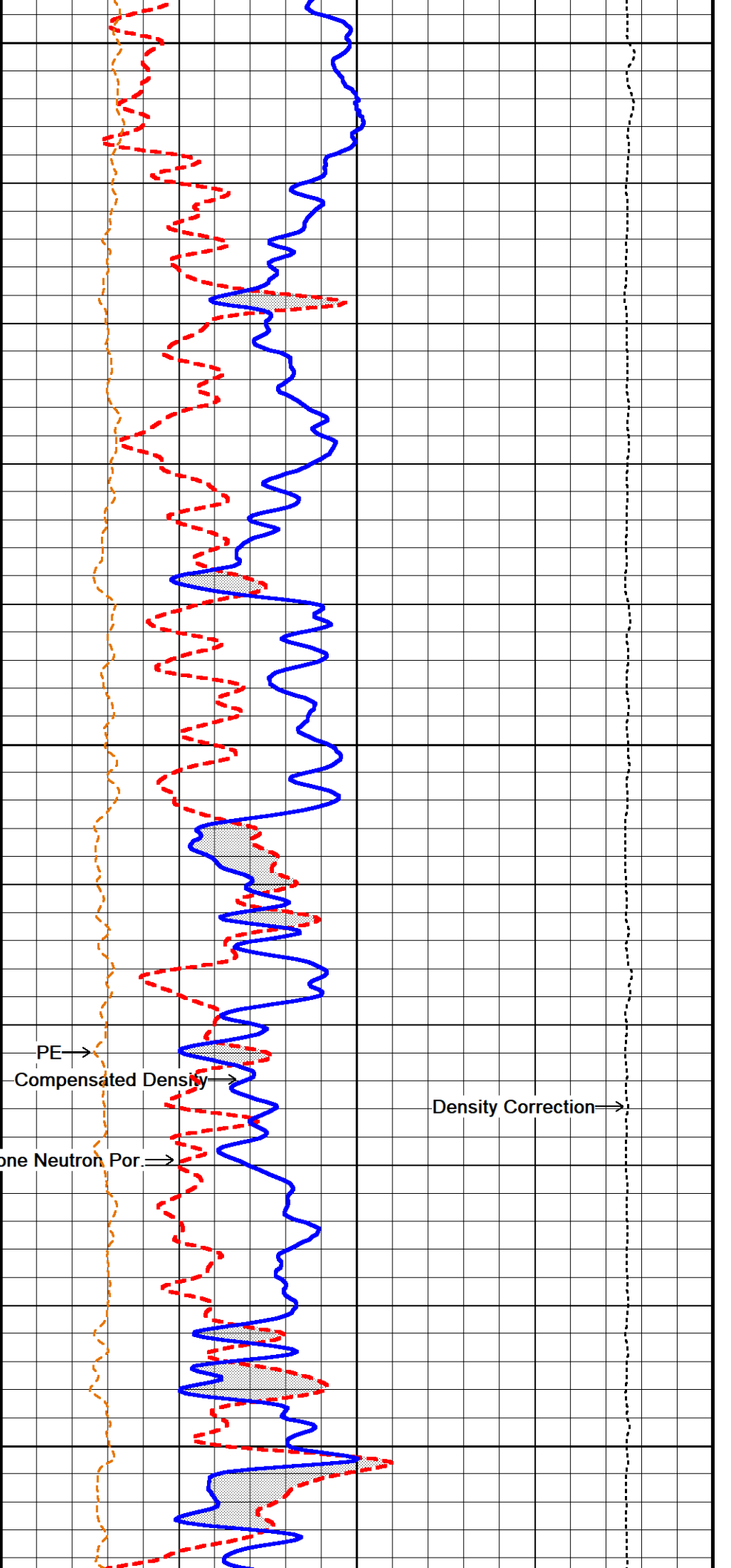
← Bit Size

← Density Caliper

← Casing Collar Locator

← Gamma Ray

DST Uphole Tension →

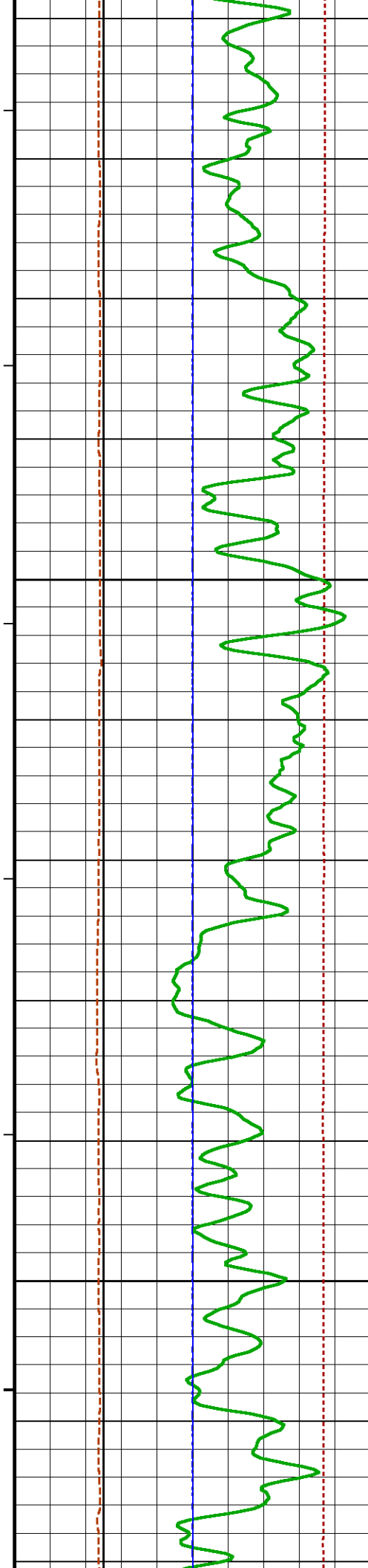


PE →

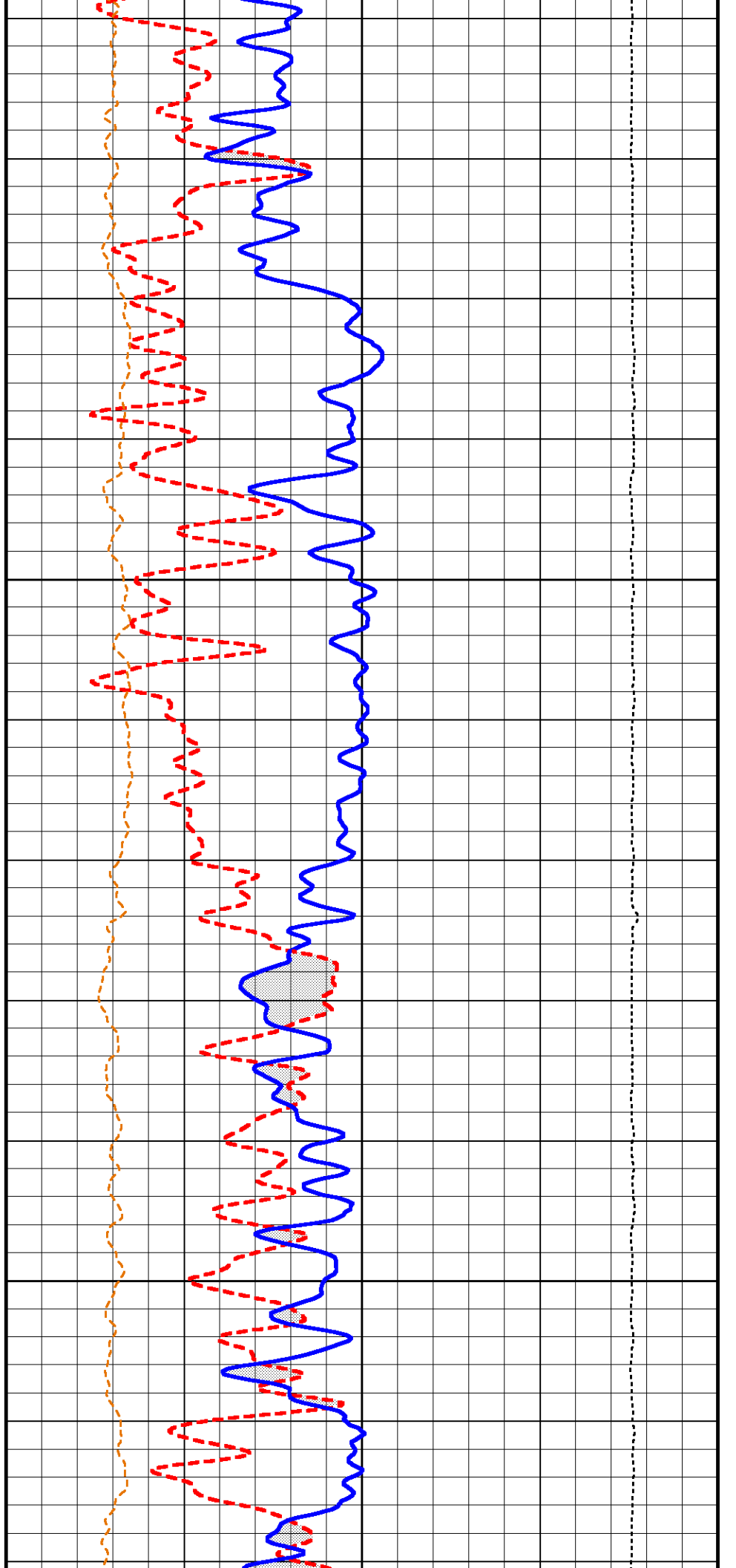
Compensated Density →

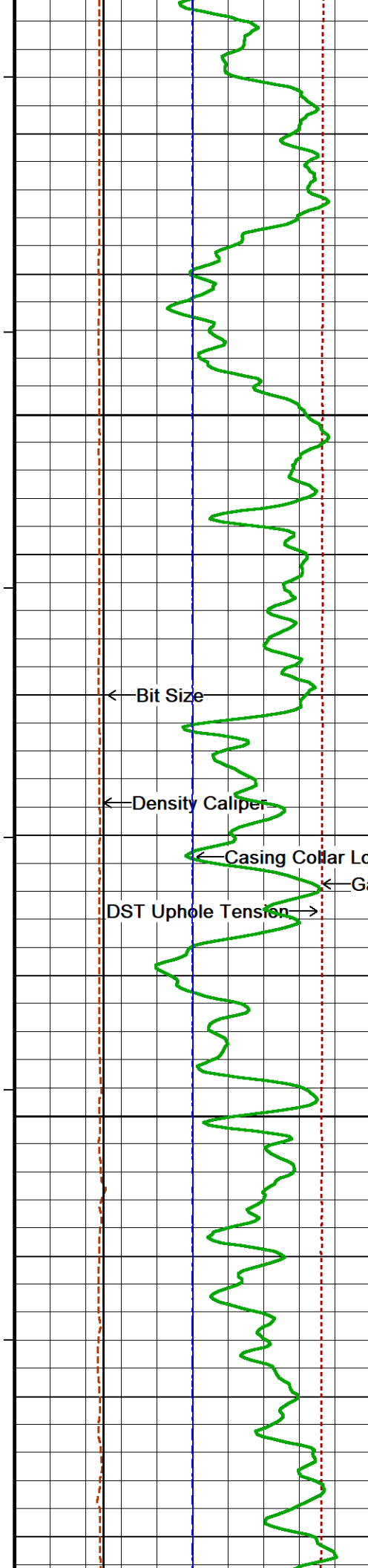
Limestone Neutron Por. →

Density Correction →



930
37°
940
37°
950
37°
960
37°
970
37°
980





37°

990

37°

1000

37°

1010

38°

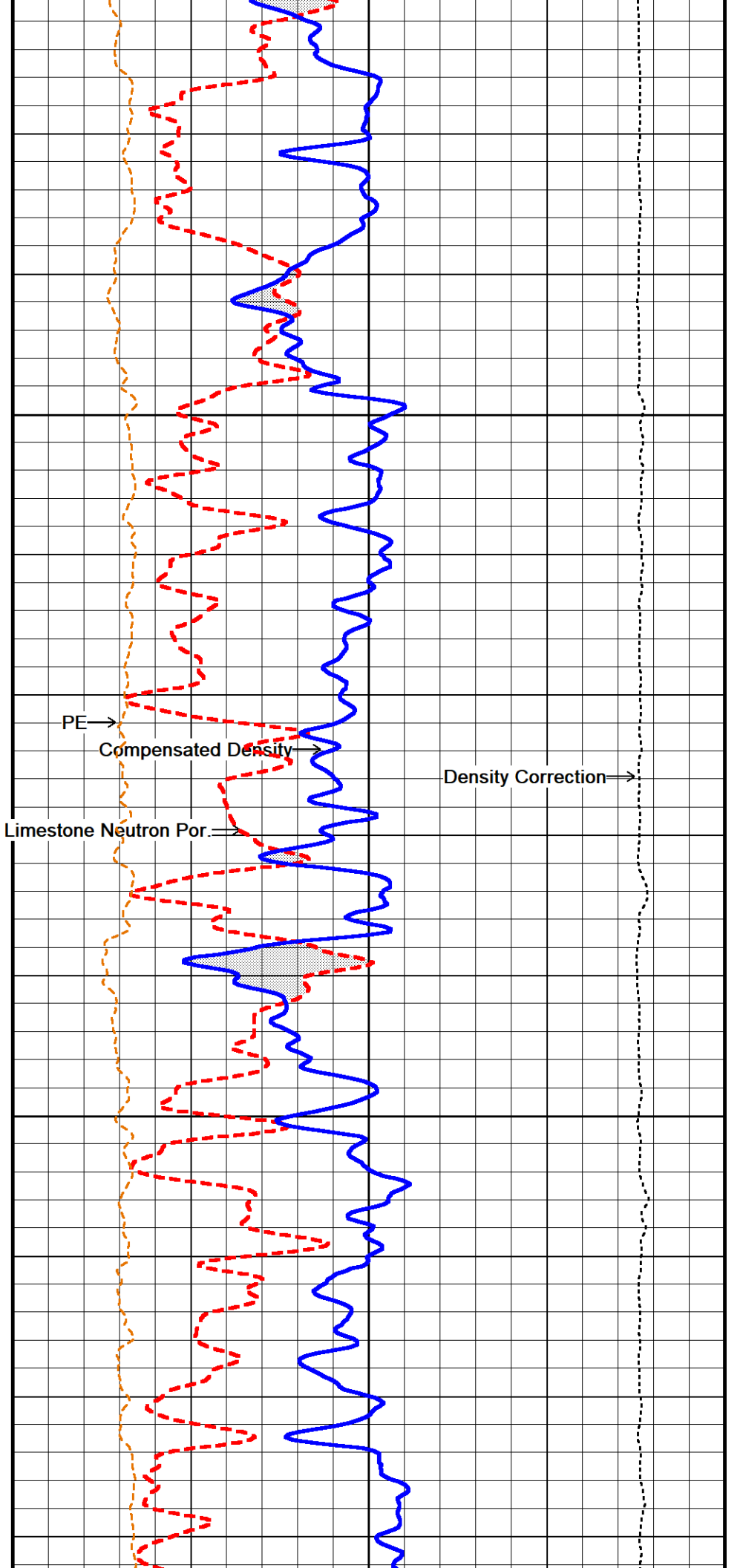
1020

38°

1030

38°

1040

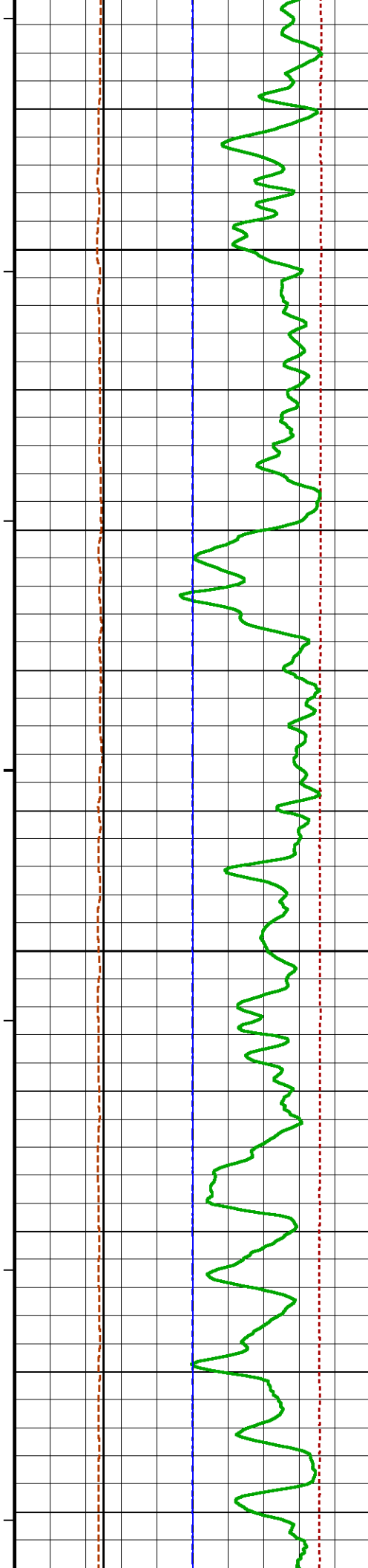


PE

Compensated Density

Limestone Neutron Por.

Density Correction



38°

1050

38°

1060

38°

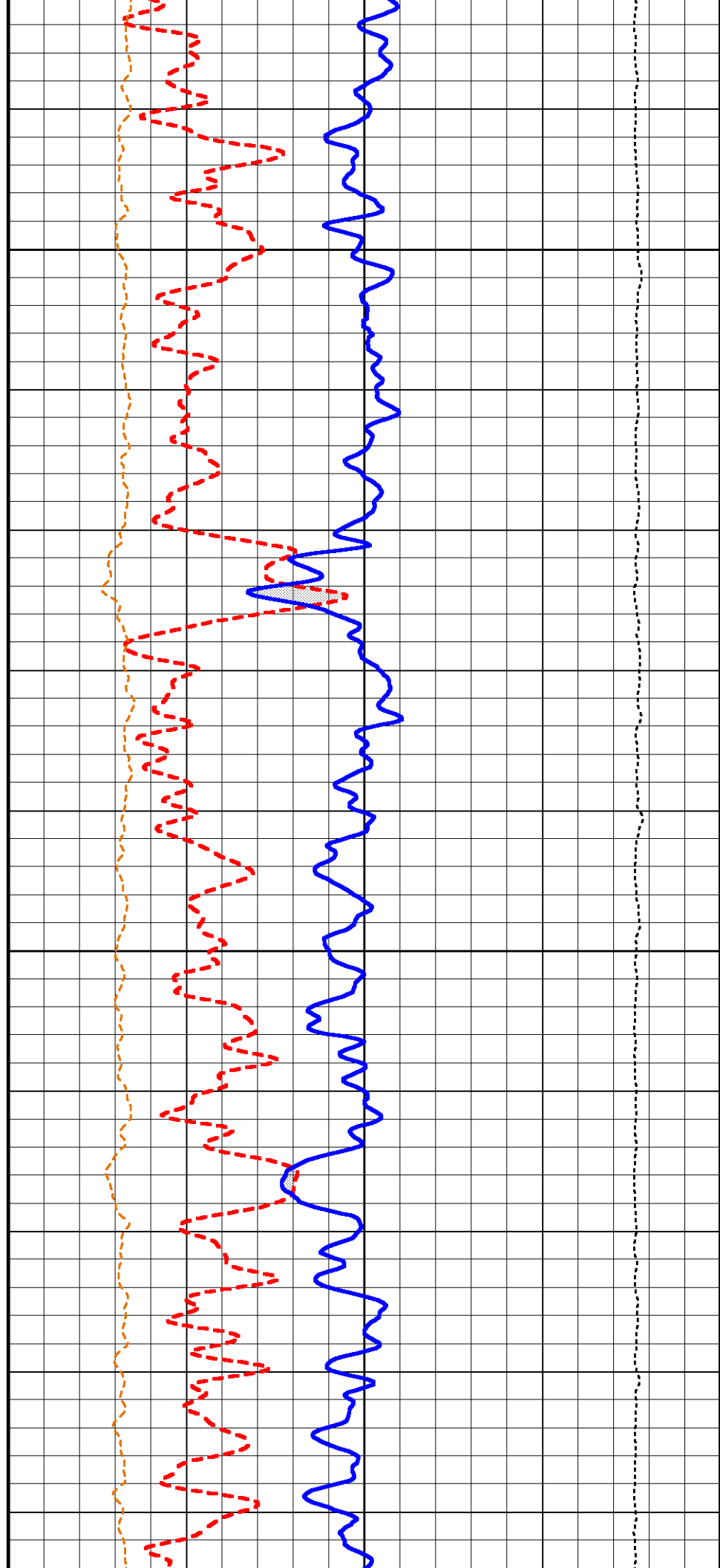
1070

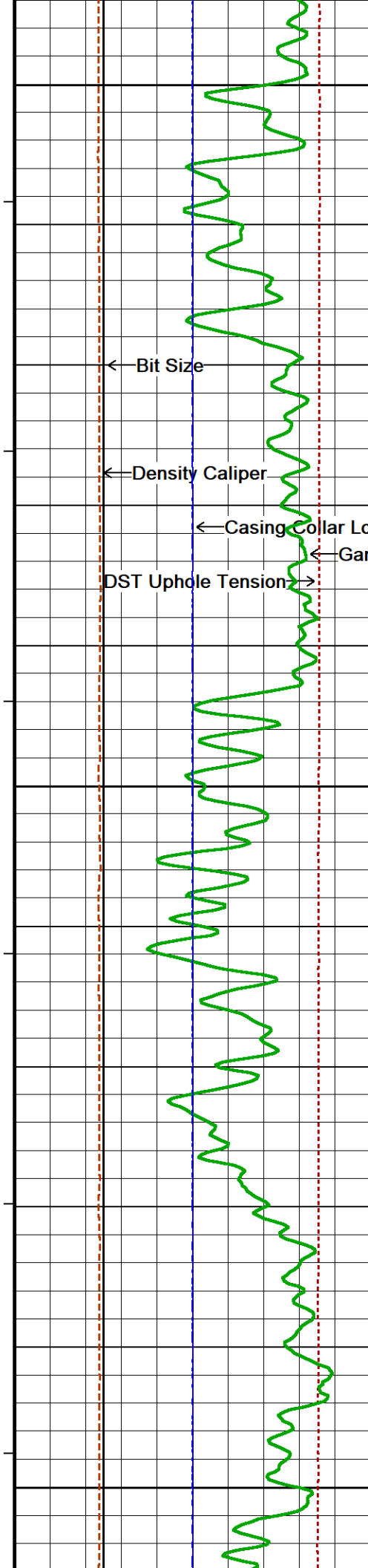
38°

1080

38°

1090





39°

1100

39°

1110

39°

1120

39°

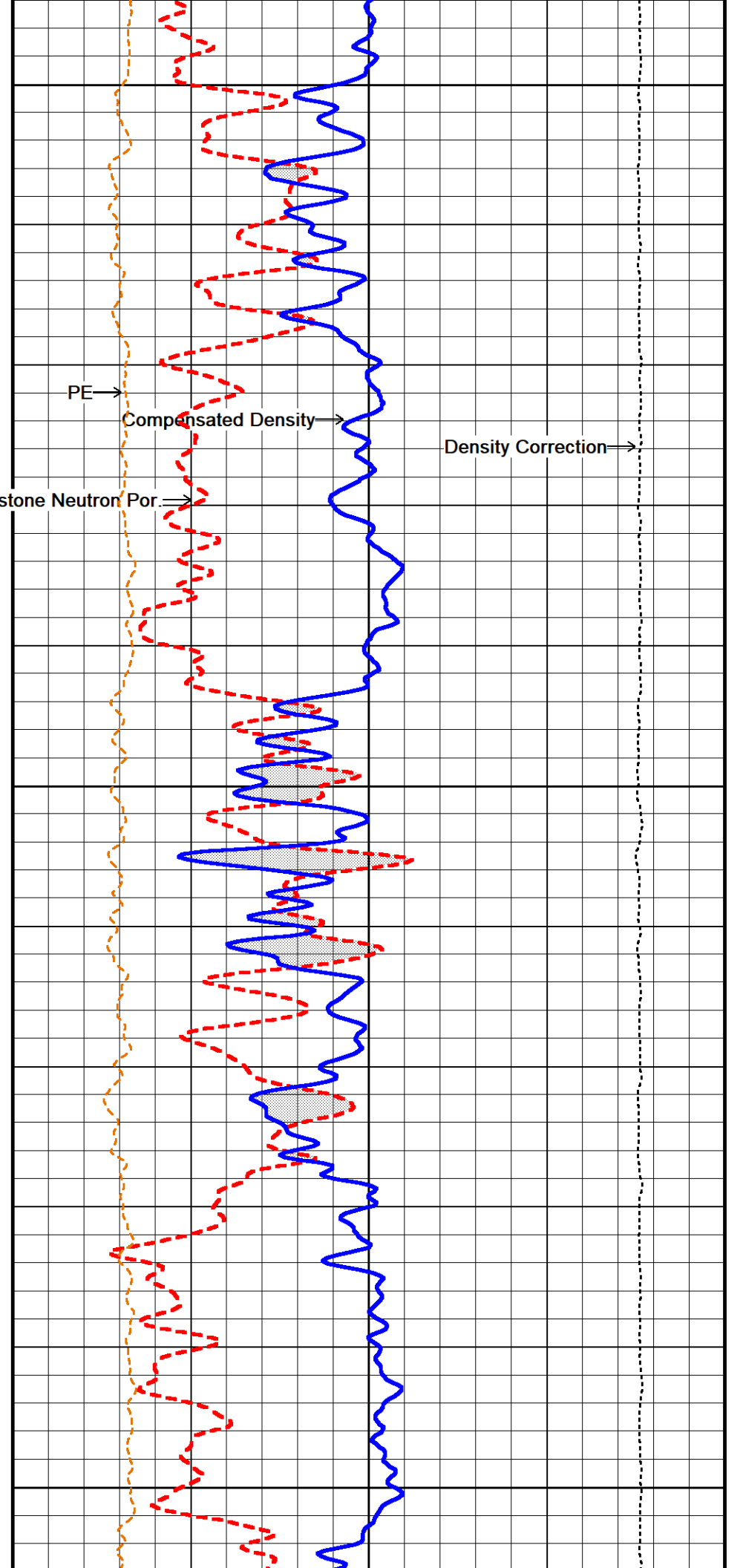
1130

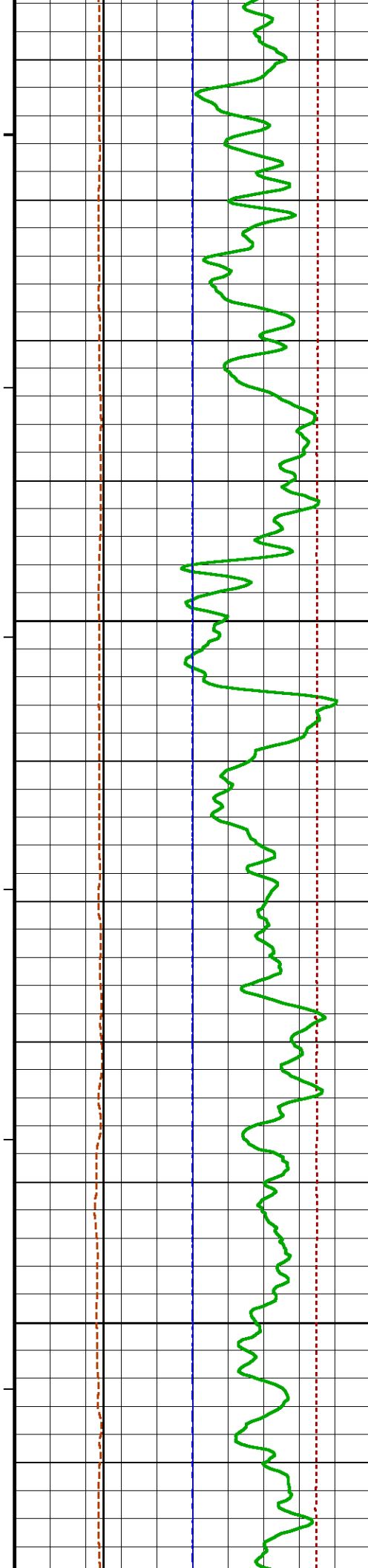
39°

1140

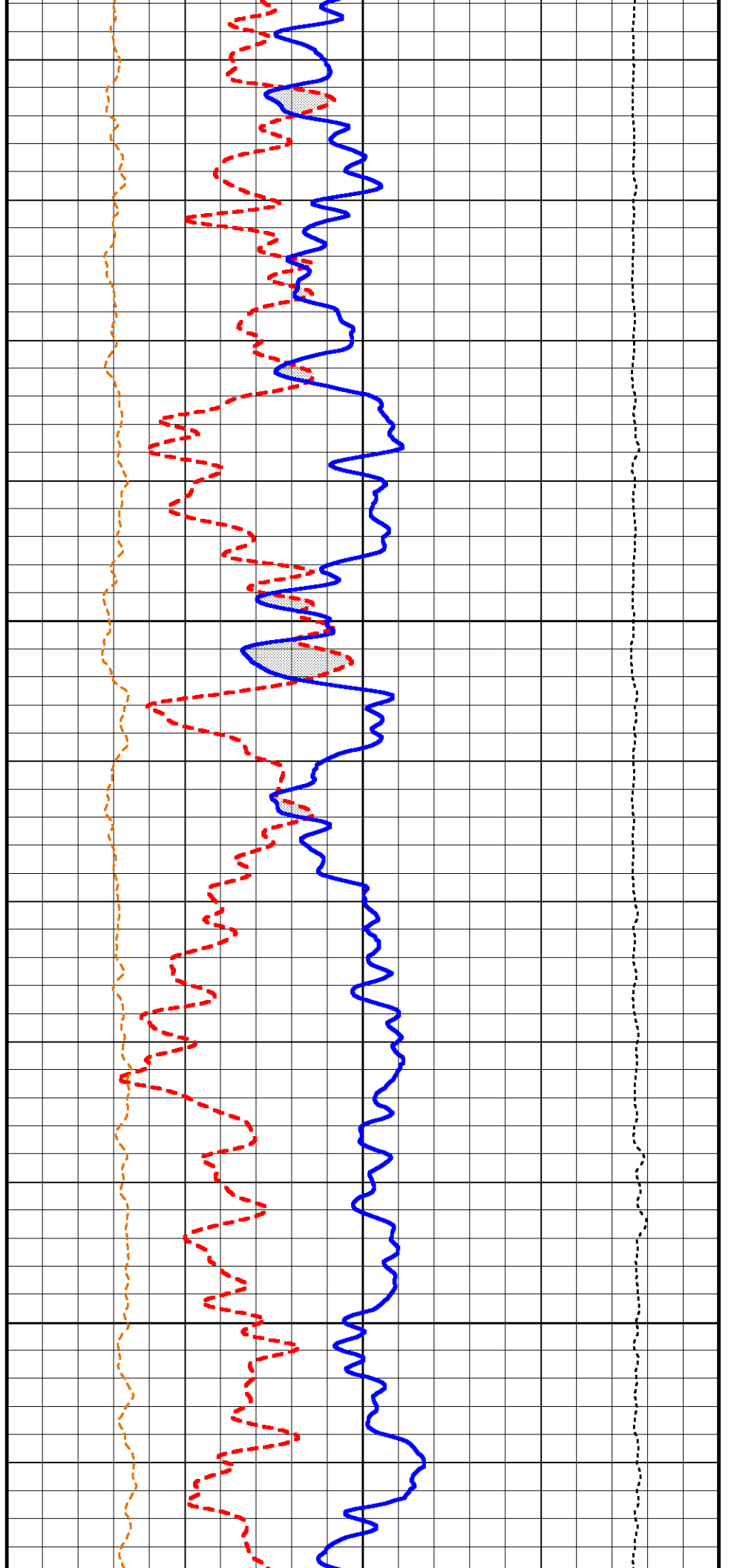
39°

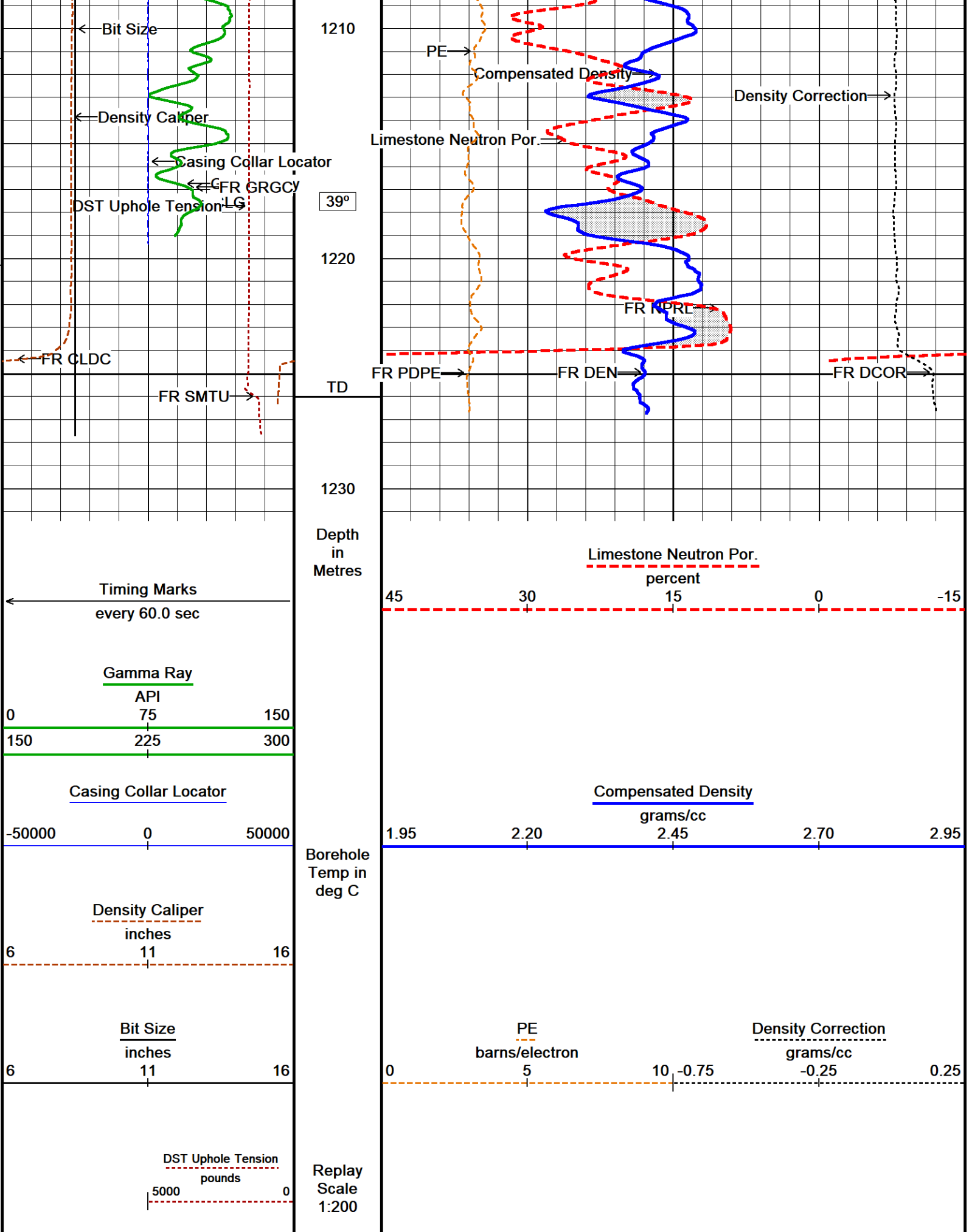
1150





39°
1160
39°
1170
39°
1180
39°
1190
39°
1200
39°





Timing Marks
every 60.0 sec

Gamma Ray
API
0 75 150
150 225 300

Casing Collar Locator
-50000 0 50000

Density Caliper
inches
6 11 16

Bit Size
inches
6 11 16

DST Uphole Tension
pounds
5000 0

Depth in Metres
1150
39°
1160

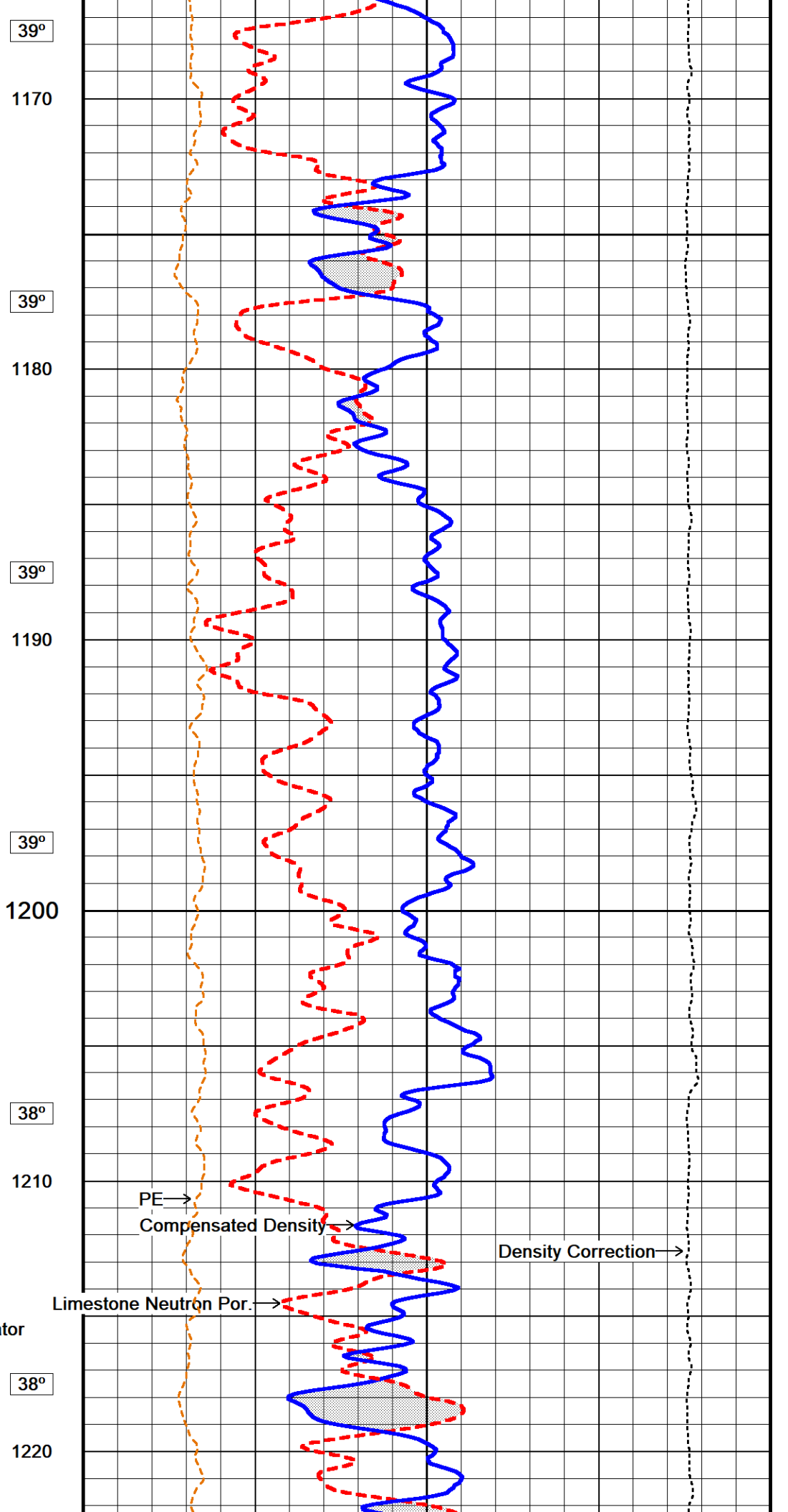
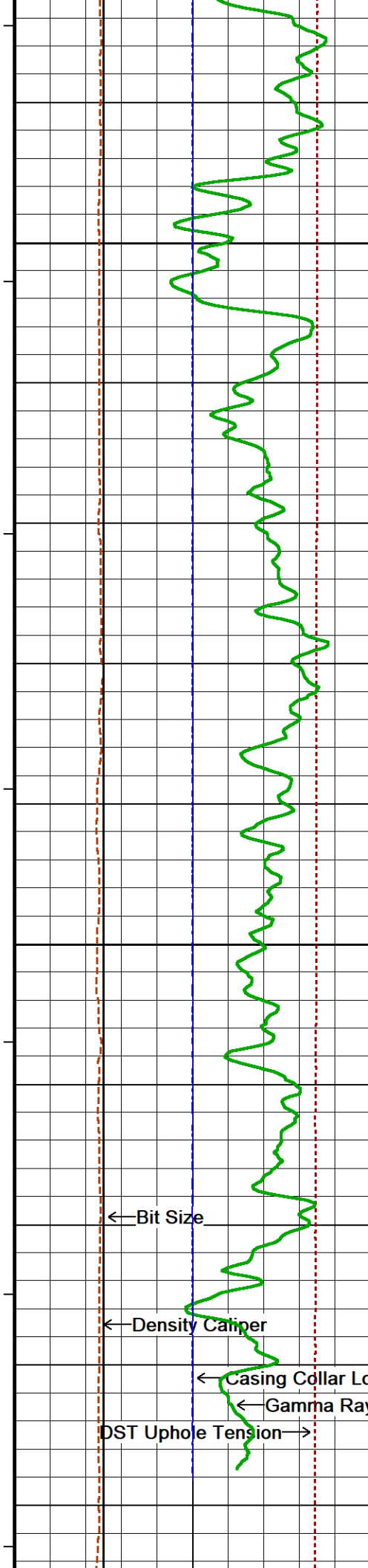
Limestone Neutron Por.
percent
45 30 15 0 -15

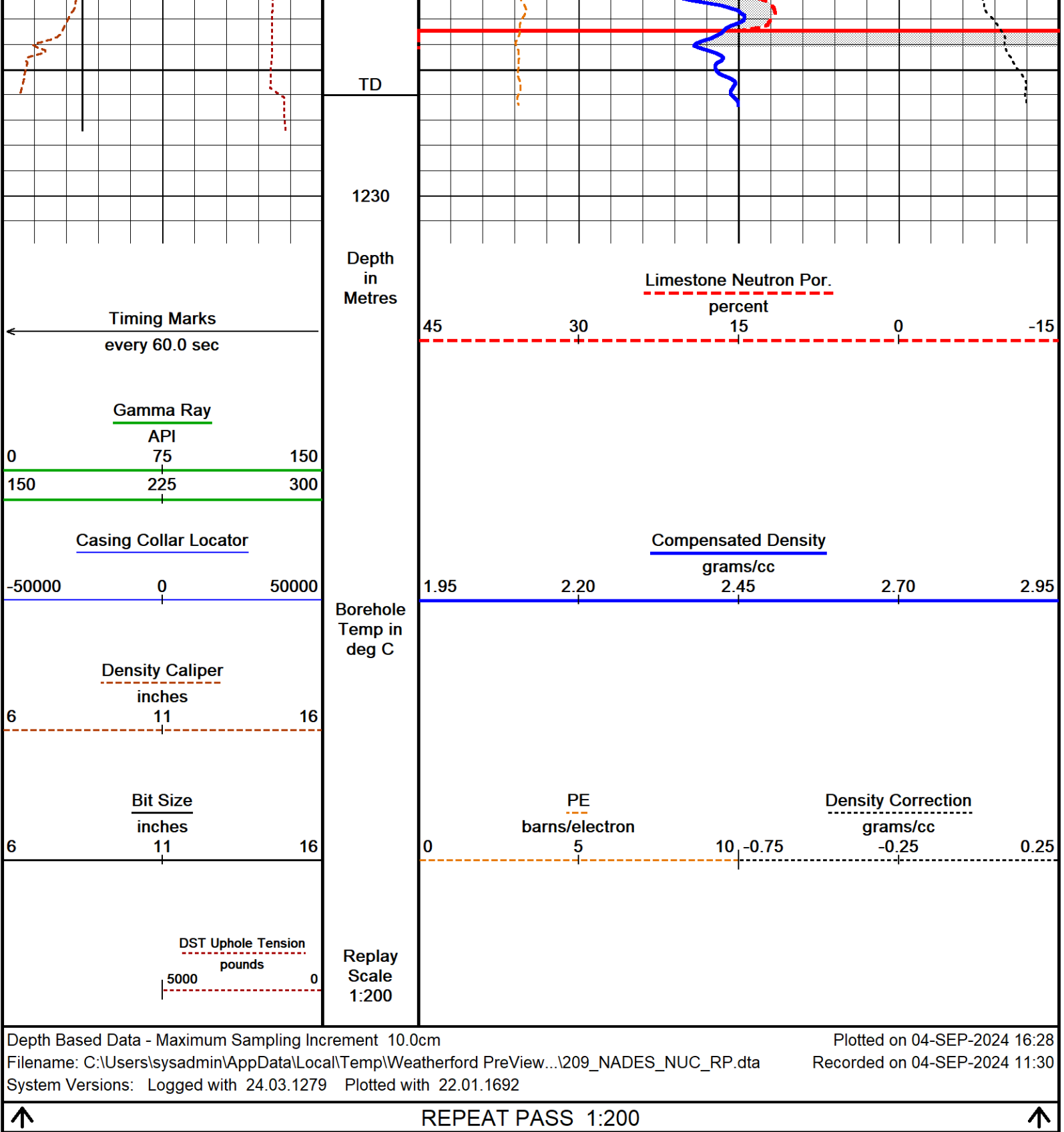
Compensated Density
grams/cc
1.95 2.20 2.45 2.70 2.95

PE
barns/electron
0 5 10

Density Correction
grams/cc
-0.75 -0.25 0.25

Replay Scale
1:200





BEFORE SURVEY CALIBRATION		
C:\Users\sysadmin\AppData\Local\Temp\Weatherford PreView8e\0\209_NADES_NUC_CALB.dta		
General Constants All 000		Last Edited on 04-SEP-2024,10:25
General Parameters		
Mud Resistivity	0.150	ohm-metres
Mud Resistivity Temperature	25.000	degrees C
Water Level	0.000	metres
Borehole Fluid Processing	Water Level Switch	
Hole/Annular Volume and Differential Caliper Parameters		

HVOL Method		Single Caliper	
HVOL Caliper 1		Density Caliper	
HVOL Caliper 2		N/A	
Annular Volume Diameter		5.500	inches
Caliper for Differential Caliper		None	
Rwa Parameters			
Porosity used		N/A	
Resistivity used		N/A	
RWA Constant A		N/A	
RWA Constant M		N/A	
SW/APOR Tool Source			
High Resolution Temperature Calibration MCG-E.A 531			
		Field Calibration on 03-SEP-2024,09:41	
	Measured	Calibrated(Deg C)	
Lower	0.00	0.00	
Upper	100.00	100.00	
High Resolution Temperature Constants MCG-E.A 531			
		Last Edited on 03-SEP-2024,09:41	
Pre-filter Length		11	
Gamma Calibration MCG-E.A 531			
		Field Calibration on 04-SEP-2024 05:54	
	Measured	Calibrated (API)	
Background	70	48	
Calibrator (Gross)	1092	755	
Calibrator (Net)	1022	707	
Gamma Calibration Tolerances MCG-E.A 531			
Ratio	1.446	1.40 1.475 1.55	Counts/API
Gamma Constants MCG-E.A 531			
		Last Edited on 04-SEP-2024,10:25	
Gamma Calibrator Number		159	
GRC-M Calibrator Jig in Use?		NO	
Inactive Background Jig in Use?		NO	
Mud Density		1.10	gm/cc
Caliper Source for Processing		Density Caliper	
Tool Position		Eccentred	
Potassium Equivalence		Chloride	
K Mud Concentration		0.00	%
Neutron Calibration MDN-C.A 528			
		Base Calibration on 02-SEP-2024 11:28	
		Field Check on 04-SEP-2024 06:06	
Base Calibration			
	Measured	Calibrated (cps)	
	Near Far	Near Far	
	2767 86	3714 110	
Ratio	32.206	33.764	
Field Calibrator at Base		Calibrated (cps)	
		2305 3386	
Ratio		0.681	
Field Check		Calibrated (cps)	
		2325 3371	
Ratio		0.690	
Neutron Calibration Tolerances MDN-C.A 528			
Ratio	32.206	-5% 33 +5%	
Base Check	0.681	0.65 0.7 0.75	
Field Check	0.690	0.661 0.681 0.701	
Neutron Constants MDN-C.A 528			
		Last Edited on 04-SEP-2024,14:15	
Neutron Source Id		P44381B	

Neutron Jig Number	131	
Air Hole Processing	Legacy	
Caliper Source for Processing	Density Caliper	
Stand-off	0.00	inches
Mud Density	1.10	gm/cc
Limestone Sigma	7.10	cu
Sandstone Sigma	4.26	cu
Dolomite Sigma	4.70	cu
Formation Pressure Source	None	
Formation Pressure	N/A	kpsi
Temperature Source	MCG External Temperature	
Temperature	N/A	degrees C
Mud Salinity	50.60	kppm
Salinity Correction	Applied	
Formation Fluid Salinity Source	None	
Formation Fluid Salinity	N/A	kppm
Barite Mud Correction	Not Applied	

Photo Density Calibration MPD-D.A 425			Base Calibration on 02-SEP-2024 11:00	
			Field Check on 04-SEP-2024 06:02	
Density Calibration				
Base Calibration		Measured	Calibrated (sdu)	
	Near	Far	Near	Far
Background	941	1089		
Reference 1	44942	20941	59082	30398
Reference 2	17725	2000	25324	2540
Field Check at Base				
	940.5	1089.1		
Field Check				
	939.1	1084.0		
PE Calibration				
Base Calibration		Measured	Calibrated	
	WS	WH	Ratio	Ratio
Background	177	828		
Reference 1	16912	44787	0.381	0.337
Reference 2	5309	17605	0.306	0.275
Field Check at Base				
	176.7	827.9		
Field Check				
	177.0	825.4		

Photo Density Calibration Tolerances MPD-D.A 425				
Near Density Ratio	2.62	-5% 2.52 +5%	Far Density Ratio	21.79 -5% 21.00 +5%
PE Calibration	0.103	0.089 0.110 0.131		
Near Den. Field Check	939.1	-3% 940.5 +3%	Far Den. Field Check	1084.0 -3% 1089.1 +3%
PE WS Field Check	177.0	-6% 176.7 +6%	PE WH Field Check	825.4 -6% 827.9 +6%

Density Constants MPD-D.A 425		Last Edited on 04-SEP-2024,11:14	
Density Source Id	P44259B		
Nylon Calibrator Number	DNCE746		
Aluminium Calibrator Number	DACD746		
Density Shoe Profile	8 inch		
Caliper Source for Processing	Density Caliper		
PE Correction to Density	Not Applied		
Mud Density	1.10	gm/cc	
Mud Density Type	Non-Barite		
Mud Filtrate Density	1.00	gm/cc	
Dry Hole Mud Filtrate Density	1.00	gm/cc	
DNCT	0.00	gm/cc	
CRCT	0.00	gm/cc	
Density Z/A Correction	Hybrid		

Precision Enhanced Density Processing	Not Applied
Density Detector Type	Compensated Density
Matrix density (gm/cc)	Depth (m)
2.71	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00
0.00	0.00

Caliper Calibration MPD-D.A 425			Base Calibration on 02-SEP-2024 10:44
			Field Calibration on 04-SEP-2024 05:57
Base Calibration			
Reading No	Measured	Calibrator Size (in)	
1	16184	3.99	
2	26146	5.96	
3	36281	7.96	
4	46147	9.85	
5	57036	11.88	
6	N/A	N/A	
Field Calibration			
	Measured Caliper (in)	Actual Caliper (in)	
	7.96	7.96	

Caliper Calibration Tolerances MPD-D.A 425			
Short Arm Field Cal.	7.96	7.76 7.96 8.16	in

DOWNHOLE EQUIPMENT

C:\Users\sysadmin\AppData\Local\Temp\Weatherford PreView8e\0\209_NADES_NUC_CALB.dta

Cablehead, 11 pin
CBH-CC 401 LG: 0.73 m WT: 24.3 lb OD: 2.240 in

11C-11B Compact Tool Adaptor
MTA-K.B 309 LG: 0.47 m WT: 13.2 lb OD: 2.240 in

Compact Swivel Head Adaptor
SHA-J.B 612 LG: 0.70 m WT: 22.0 lb OD: 2.244 in

Compact Comms Gamma
MCG-E.A 531 LG: 2.65 m WT: 63.9 lb OD: 2.244 in

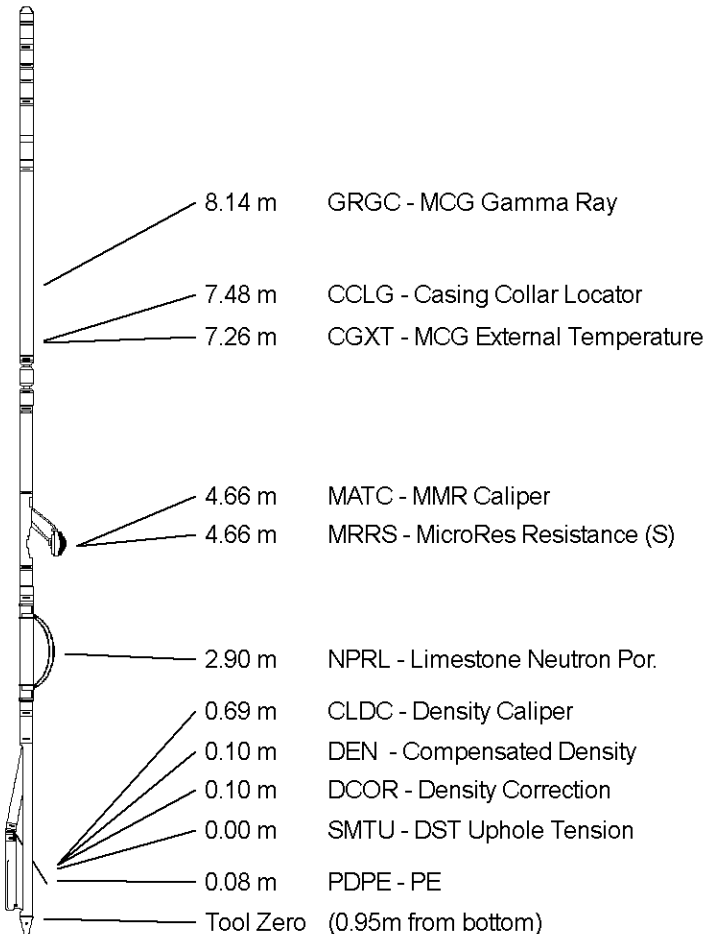
Compact Knuckle Joint
SKJ-E.A 255 LG: 0.66 m WT: 24.3 lb OD: 2.240 in

Compact Micro-Resistivity
MMR-C.A 219 LG: 2.62 m WT: 81.6 lb OD: 3.819 in

Compact Neutron
MDN-C.A 528 LG: 1.53 m WT: 50.7 lb OD: 2.244 in

Compact Density/Caliper
MPD-D.A 425 LG: 2.92 m WT: 90.4 lb OD: 2.449 in

Pressure Bung + Hole Finder
HFS 3 LG: 0.28 m WT: 6.6 lb OD: 2.244 in



All measurements relative to tool zero.

Total Length: 12.56 m Weight: 377.0 lb

COMPANY	S.N.G.N. ROMGAZ S.A.
WELL	209 NADES
FIELD	NADES
PROVINCE/COUNTY	MURES
COUNTRY/STATE	ROMANIA

Elevation Kelly Bushing	484.71	metres	Last Reading	250.00	metres
Elevation Drill Floor	484.71	metres	First Reading	1225.00	metres
Elevation Ground Level	480.21	metres	Depth Driller	1225.00	metres
			Depth Logger	1226.00	metres



Weatherford®

PHOTO DENSITY
DUAL NEUTRON